

60°N Signal Works Audio Bench Suite Handbook

macOS edition

Applications: Matrix Bench, MIDI Bench, Signal Bench, Spectral Bench, Latency Bench

Purpose: User handbook and measurement guide for the Audio Bench Suite.

This publication edition documents the current qualified macOS releases of the Audio Bench Suite. Windows and Linux are mentioned only where platform context is technically useful. The handbook combines operating guidance, measurement methodology, validation evidence and technical reference material, with the final application screenshots and technical diagrams placed in context. Screenshots show the qualified application states used for this edition, while diagrams clarify physical routing and cross-Bench measurement workflows.

Table of contents

Chapter page numbers below correspond to this publication edition.

1. Audio Bench Suite.....	2
2. Audio and measurement fundamentals.....	4
3. Signal Bench.....	7
4. Spectral Bench.....	17
5. Latency Bench.....	26
6. Matrix Bench.....	35
7. MIDI Bench.....	45
8. Cross-Bench measurement workflows.....	53
9. Measurement confidence and validation.....	61
10. Troubleshooting and measurement pitfalls.....	63
11. Technical reference.....	74
12. Bugs, errata and current packages.....	83

How to use this handbook

The handbook is deliberately layered. For an immediate measurement, go directly to the relevant Bench chapter. Chapters 2, 8, 9 and 10 provide the shared measurement concepts, cross-Bench workflows, confidence guidance and troubleshooting needed when interpretation matters.

Application chapters focus on operating the tools and interpreting their results. Internal development and maintenance details are intentionally kept outside this user handbook unless they directly affect measurement behavior.

Which Bench should I use?

Goal	Primary Bench	Often useful with
------	---------------	-------------------

Generate tones, noise, two-tone/IM and other test stimuli	Signal Bench	Spectral Bench
Inspect spectrum, level, harmonics, THD/THD+N, IMD, sweeps and phase	Spectral Bench	Signal Bench
Measure physical-path or DUT latency	Latency Bench	Matrix Bench / external routing
Route, split, mix, process and monitor physical and virtual audio	Matrix Bench	All audio Benches
Monitor, transmit and sequence MIDI	MIDI Bench	MIDI-controlled DUTs

Measurement mindset

The Bench applications are tools, not substitutes for defining the measurement. A numerically precise result can still answer the wrong question if the signal path, level, sample rate, routing or measurement definition is wrong.

1. **Define what is being measured.** Transport latency, complete filtered-response timing, THD, THD+N, IMD and a spectrum trace answer different questions.
2. **Control the signal path.** Avoid clipping, unintended feedback, wrong channels, accidental sample-rate conversion and processing that is not part of the intended DUT path.
3. **Establish a reference.** A cable loopback, baseline, known signal or other reference separates the DUT from the measurement setup where the method allows it.
4. **Check repeatability.** Repeated measurements expose unstable routing, clocking, signal conditions and ambiguous estimators.
5. **Record enough context to reproduce the result.** Application version, device, sample rate, channel routing, relevant buffer settings, DUT state and measurement settings all matter.

1. Audio Bench Suite

1.1 Purpose and scope

The 60°N Signal Works Audio Bench Suite is a collection of focused macOS engineering tools rather than one monolithic measurement application. Each Bench has a narrow primary responsibility, and the applications can be combined when a measurement needs generation, routing, analysis, timing or MIDI control at the same time.

The suite currently consists of:

- **Signal Bench**, deterministic audio test-signal generation.
- **Spectral Bench**, spectral, level, distortion, sweep and phase analysis.
- **Latency Bench**, physical-path and DUT latency measurement.
- **Matrix Bench**, low-latency physical and virtual audio routing, mixing, monitoring and utility processing.
- **MIDI Bench**, MIDI monitoring, transmission and deterministic command-file sequencing.

The handbook documents the current qualified macOS release of each application. Other versions or platforms are mentioned only where they materially affect compatibility or measurement behavior.

1.2 Installation model

The intended macOS installation location for the applications is:

/Applications/60°N Signal Works Audio Bench Suite

Public installation packages: [Google Drive distribution folder](#). Use the package appropriate to the current documented release unless reproducing an older measurement.

Install each Bench from its current macOS package. When reproducing an older measurement, record the application version because behavior and available controls can change between releases.

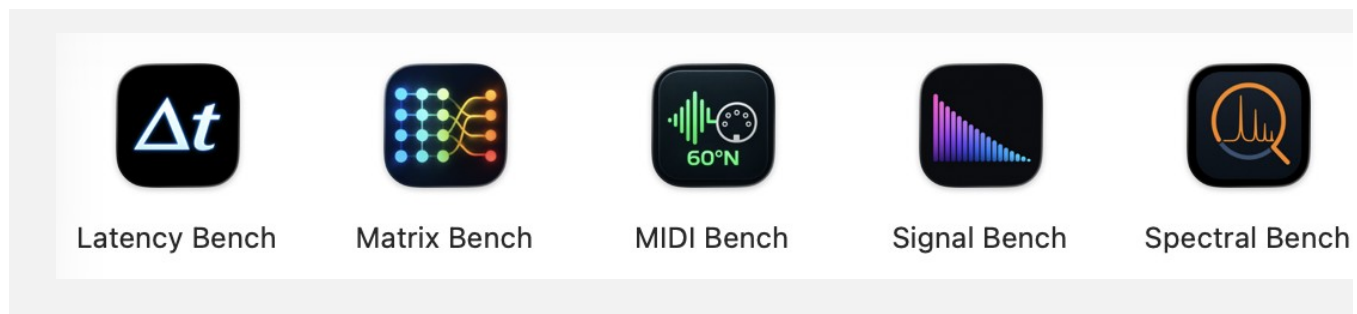


Figure 1.1. The five Audio Bench Suite applications installed on macOS.

1.3 A practical way to choose the tool

Start from the physical question rather than from the application. If the DUT needs a known stimulus, start with Signal Bench. If the unknown is spectral behavior, distortion or phase, use Spectral Bench. If the unknown is elapsed time through a physical path, use Latency Bench. Use Matrix Bench when the difficulty is routing, monitoring, splitting or combining physical and virtual paths. Use MIDI Bench when repeatable MIDI control or observation is part of the test.

A single experiment may use several Benches. For example, Signal Bench can excite a DUT while Spectral Bench observes its output. Matrix Bench can provide repeatable routing around that setup. MIDI Bench can change DUT states between measurements. Latency Bench is deliberately more self-contained because its timing estimator generates and captures its own probe.

1.4 Common setup principles

Use a known audio interface and explicit input/output channels. Set the intended sample rate before beginning a measurement. Where buffer size is relevant, choose it intentionally and record it. Disable unrelated processing unless it is deliberately part of the path under test.

For physical loopback work, label cables and channels. Verify routing at a conservative level before increasing signal level. A surprising number of apparent algorithmic or DUT failures are actually wrong-channel, wrong-loop or gain-staging errors.

Bluetooth and aggregate or virtual routing deserve extra attention because they can introduce fixed sample-rate constraints, additional buffering, sample-rate conversion or independent clock domains. Those effects are not automatically measurement errors; they become errors when they are present but not part of the quantity the test is meant to measure.

1.5 Reproducibility record

For measurements worth keeping, record at least:

- Bench name and version;
- macOS audio device and relevant input/output channels;
- sample rate;
- buffer size where it can affect the result;
- DUT identity and state;
- physical and virtual routing;
- generator and analyzer settings;
- baseline/reference state where applicable;
- repeated-run statistics or other evidence of stability when the tool provides them.

Saved result files should be kept with enough human-readable context that they still make sense after the physical setup has been dismantled.

2. Audio and measurement fundamentals

This chapter establishes the conventions used throughout the handbook. The individual Bench chapters provide application-specific detail, but the same distinctions between digital level, analog level, samples, time, phase and physical signal paths apply everywhere.

2.1 Digital level and dBFS

The audio Benches use digital full scale as the primary digital amplitude reference. 0 dBFS is the maximum representable reference level of the digital system; it is not an analog voltage. A setting of, for example, -20 dBFS says nothing by itself about volts at an interface output or volts arriving at a DUT input. The analog value also depends on interface calibration, output gain, input sensitivity and any intervening analog gain or attenuation.

Peak, RMS and spectral level are also different quantities. A sine wave's peak level, its RMS level and the amplitude reported for its FFT component are related only when the measurement convention is known. Likewise, broadband-noise RMS cannot be compared directly with the level of one FFT bin without accounting for bandwidth and analyzer normalization.

The practical rule is simple: compare like with like, and state the level convention when the distinction matters.

2.2 Headroom, clipping and gain staging

Digital clipping occurs when a signal exceeds the representable range at a point in the digital path. Analog clipping can occur earlier or later in the chain even while the digital meter appears safe. Conversely, a signal can be so low that analog noise, converter noise or numerical uncertainty dominates the measurement.

For test work, begin conservatively, verify the path, then raise the stimulus only as far as needed for useful signal-to-noise ratio and DUT operating level. If distortion is being measured, distinguish distortion intentionally produced by the DUT from clipping accidentally produced by the generator, interface or analyzer input.

2.3 Sample rate, samples and time

A sampled system represents time in discrete intervals. At sample rate F_s , one sample corresponds to: $\text{sample time} = 1 / F_s$

At 48 kHz, one sample is approximately 20.833 μs ; 48 samples equal 1 ms. At 44.1 kHz, one sample is approximately 22.676 μs .

Latency Bench can report fractional-sample timing because its correlation peak is interpolated rather than restricted to an integer sample index. Fractional-sample reporting does not imply that the audio interface physically delays data by a fractional hardware sample; it is an estimate of relative timing from the captured waveform.

2.4 Buffer size is not the same as latency

Audio buffer size affects scheduling and can be a major component of end-to-end latency, but it is not itself a complete latency specification. Hardware safety buffers, converter filters, USB or other transport buffering, driver behavior, application callbacks, independent clock domains and sample-rate conversion can all contribute.

The same distinction matters in the opposite direction: a client callback size does not necessarily reveal every internal hardware or driver buffer in the path. When latency matters, measure the path that matters rather than calculating a result from one buffer-size field alone.

2.5 Frequency, bandwidth and spectral resolution

Frequency describes periodic rate; bandwidth describes a range of frequencies. In FFT analysis, the nominal bin spacing is determined by sample rate and FFT length. A larger FFT provides finer bin spacing but represents a longer time record. Windowing changes spectral leakage and the effective noise bandwidth, so bin spacing alone does not describe the analyzer's ability to separate or quantify arbitrary components.

For broadband measurements, bandwidth is part of the result. Noise integrated over a wider bandwidth normally contains more total power than the same noise process integrated over a narrower bandwidth. This is one reason THD and THD+N must not be treated as interchangeable measurements.

2.6 FFT windows, coherent gain and noise bandwidth

An FFT operates on a finite block. Unless the captured waveform joins perfectly at the block boundaries, treating that block as periodic produces leakage. A window weights the block to control that leakage, but the weighting changes amplitude and noise behavior.

Two corrections therefore matter in a calibrated analyzer:

- **coherent gain**, which relates the windowed FFT amplitude of a coherent tone to its true amplitude;
- **equivalent noise bandwidth (ENBW)**, which describes the noise bandwidth represented by a windowed FFT bin.

Spectral Bench defines its own spectrum and measurement semantics in detail. Do not infer calibrated tone or noise values from an arbitrary FFT display without knowing these conventions.

2.7 Frequency response, phase and polarity

Frequency response is complex: it has both magnitude and phase. A filter can leave a frequency's magnitude nearly unchanged while rotating its phase, or strongly change both.

Polarity is a separate concept. Reversing polarity multiplies the waveform by -1, equivalent to a 180-degree phase inversion at every frequency in the ideal linear case. A constant polarity reversal is therefore not the same thing as frequency-dependent phase shift.

This distinction matters in both spectral and timing work. A correlation method that uses correlation magnitude can still identify a time offset through an inverted path, while preserving the sign of the raw correlation as useful evidence about polarity. The application-specific estimator rules are documented in the Latency Bench chapter.

2.8 Phase delay and group delay

A causal filter necessarily has phase behavior associated with its transfer function. Group delay describes how phase slope varies with frequency and is especially important for bandwidth-limited signals, crossovers and steep filters.

This leads to a critical interpretation rule for latency measurements: the timing of a **complete filtered response** is not necessarily the same quantity as the DUT's bare digital transport latency. A steep HPF, LPF, crossover or band-pass can shift the dominant correlation timing by several milliseconds even if the underlying transport delay is unchanged.

Neither result is inherently wrong. They answer different questions. The measurement setup and report must make clear which quantity is being interpreted.

2.9 Correlation and time-of-arrival estimation

Cross-correlation compares two signals while shifting one relative to the other. A peak indicates the displacement at which the signals are most alike according to the estimator. With a broadband deterministic probe, this can provide precise relative timing even when the DUT changes level or introduces moderate linear filtering.

The highest numerical peak is not automatically trustworthy. Strong filtering can broaden a correlation lobe, reduce overlap, or create several plausible peaks. A robust estimator therefore needs evidence and ambiguity rules in addition to peak finding. Latency Bench deliberately rejects measurements when competing timing interpretations are too close rather than forcing every capture into a result.

2.10 Repeatability, accuracy and uncertainty

These terms answer different questions. **Repeatability** describes how closely repeated measurements agree under the same conditions. **Accuracy** describes closeness to the quantity intended to be measured.

A result can be extremely repeatable yet systematically wrong because the baseline, routing or interpretation is wrong.

Measurement uncertainty includes more than statistical run-to-run variation. Device clocks, analog noise, estimator behavior, filter phase, sample-rate conversion, channel mismatch and setup changes can all matter. Where a Bench reports standard deviation or competing candidates, treat those as evidence about a specific part of the measurement, not as a complete uncertainty budget for the entire physical experiment.

2.11 Baselines and reference paths

A baseline removes a known contribution only if the baseline represents the contribution that should be removed. In a physical latency measurement, a direct reference loop can characterize interface and routing delay so that the DUT measurement can be expressed relative to it. If cables, devices, sample rate or channel assignments change, the old baseline may no longer describe the current setup.

The same principle applies beyond latency. A known generator level, loopback spectrum or reference device can expose measurement-system behavior before an unknown DUT is introduced.

2.12 Physical paths, virtual paths and clock domains

A path entirely inside one synchronous digital clock domain behaves differently from a path crossing independent devices. When two hardware devices run from independent clocks, their nominal sample rates are not exactly identical. Long-running routing may therefore require asynchronous sample-rate conversion or another clock-domain strategy.

Virtual audio devices add routing flexibility but do not make clocking, buffering or feedback concerns disappear. Matrix Bench's virtual-device and independent-output-clock behavior is covered in its own chapter.

2.13 A minimum pre-measurement check

Before trusting an important result, confirm:

1. the intended input and output devices and channels;
2. the intended sample rate;
3. safe, useful signal levels without unintended clipping;
4. the intended DUT state and processing;
5. the reference or baseline, if the method depends on one;
6. repeatability over more than one run when practical;
7. that the reported quantity is actually the one the experiment was designed to measure.

That short check prevents more bad measurements than additional decimal places ever will.

3. Signal Bench

Signal Bench 2.0.0 for macOS is the suite's deterministic audio test-signal generator. It provides broadband Pink and White noise, single- and dual-sine stimuli, standard analyzer-test presets, guitar-oriented two-tone stimuli, deterministic envelope shaping and optional harmonic or pick-attack components.

The design priority is repeatability. Signal Bench is intended to produce known, controllable stimuli for engineering work, not to imitate an instrument or make a test signal subjectively more natural.



Figure 3.1. Signal Bench 2.0.0 main window and primary signal-generation controls.

3.1 Before generating a signal

Choose the required signal type, begin with a conservative **Output Level**, verify the intended output device and physical routing, and only then raise the level as required.

A digital level in dBFS does not define the analog voltage at a DUT. Interface calibration, output gain and any analog attenuation or amplification determine the actual voltage. Signal Bench deliberately does not insert an automatic limiter, because limiting would alter the measurement stimulus. Complex Dual Sine combinations can therefore exceed full scale if individual tone levels, harmonics, Pick Attack and master level are set too high.

The default master Output Level is -18.0 dB, providing a conservative starting point.

3.2 Signal modes at a glance

Signal Bench has three primary signal modes:

Mode	Primary use
Pink	broadband testing with approximately equal

	energy per octave
White	broadband testing with approximately equal power density per Hz
Dual Sine	single-tone, two-tone, IMD and deterministic guitar-oriented stimuli

Pink and White can use bandwidth filters. Dual Sine instead generates its spectral components directly and provides the tone, preset, harmonic, Slicer and Pick Attack controls described below.

3.3 Pink noise

Signal Bench uses Stefan Stenzel's **A New Shade of Pink** multirate algorithm, specifically the extended low-frequency form with 20 octave-spaced one-bit noise sources. The method combines octave-rate one-bit sequences using first-order-hold interpolation and a small FIR correction near Nyquist.

This architecture is attractive for a real-time generator because it combines good spectral accuracy with low computational cost. Signal Bench adapts the algorithm to ordinary C++ floating-point arithmetic rather than relying on the IEEE-754 accumulator trick used by the reference implementation.

A fixed scale factor of 1.6 follows the generator. It is a compatibility normalization for practical RMS level and headroom, not part of Stenzel's algorithm and not a change to the spectral shape.

Left and right Pink channels use independent deterministic states, so the stereo output is decorrelated rather than duplicated mono noise. Resetting the processor restores deterministic starting states, which is useful in repeatable testing.

3.3.1 Pink-noise reference checks

The implementation has been validated using a deterministic 16,777,216-sample float capture, overlapping Hann-window periodograms and 1/6-octave power averaging. Spectral shape is compared with a fitted ideal 1/f power spectral density, independently of absolute level.

Interpreted sample rate	RMS spectral error vs fitted 1/f PSD	Maximum absolute 1/6-octave-bin error
44.1 kHz	0.066 dB	0.212 dB
48 kHz	0.070 dB	0.284 dB
96 kHz	0.085 dB	0.295 dB
192 kHz	0.118 dB	0.350 dB

The same reference run measured 0.195411 RMS (-14.18 dBFS) and a peak magnitude of 0.916451.

The published acceptance limits are deliberately wider than the observed reference values: RMS 0.185-0.205, peak magnitude below 1.0, RMS spectral error no greater than 0.15 dB, and maximum 1/6-octave-

bin error no greater than 0.40 dB. These limits are useful when checking that Pink-noise behavior remains consistent across supported sample rates.

The finite reference record has a mean of approximately +0.0499. With the 20-source low-frequency form, extremely slow components evolve over timescales longer than the reference capture, so a finite record is not required to average close to zero. A true deterministic DC component would require a different test.

3.3.2 Pink-noise algorithm credit

The pink-noise algorithm is based on work by **Stefan Stenzel**, who directly suggested its use in Signal Bench. His *A New Shade of Pink / Not so new shade of pink* work describes the multirate first-order-hold method, LFSR-based one-bit noise sources and FIR lookup-table correction. Signal Bench's C++17/JUCE implementation is an adaptation rather than a verbatim copy of the reference source.

Detailed algorithm references and engineering validation material are maintained separately from this user handbook.

3.4 White noise

White noise is generated directly from internal xorshift64* pseudorandom-number generators. The two output channels use different deterministic seeds and independent states.

Unlike Pink noise, White noise is not passed through the pinking stage. The optional Low Cut and High Cut filters remain available.

3.5 Broadband bandwidth controls

Pink and White modes provide independent bandwidth controls:

Control	Range / behavior
Low Cut	10 Hz to 2 kHz
High Cut	500 Hz to 24 kHz
Slope	6, 12 or 24 dB/octave
Default	both filters Off; 12 dB/oct selected

The 6 dB/oct setting uses a first-order section, 12 dB/oct uses a second-order Butterworth section, and 24 dB/oct uses two cascaded second-order sections forming a fourth-order Butterworth alignment. High Cut is clamped internally below Nyquist when necessary for the current sample rate.

These filters are disabled in Dual Sine mode.

3.6 Dual Sine

Dual Sine generates two independently adjustable sinusoidal components.

Parameter	Range	Default
F1	10 Hz to 24 kHz	110.00 Hz

F2	10 Hz to 24 kHz	164.81378 Hz
Tone 1 Level	-60 to 0 dB	-6.0 dB
Tone 2 Level	-60 to 0 dB	-6.0 dB

The default frequencies are A2 and E3 in 12-tone equal temperament.

Each oscillator uses a phase accumulator. For frequency f , sample rate F_s and phase ϕ :

$$x = \sin(\phi)$$

$$\phi_{\text{next}} = \phi + 2\pi f / F_s$$

with phase wrapping at 2π .

Frequency changes are smoothed over approximately 20 ms. The oscillator phase itself is not reset, so a manual frequency change does not create an artificial hard phase restart. The resulting Dual Sine sample is copied coherently to every active output channel.

3.7 Single-tone, two-tone and IMD presets

A nonlinear DUT produces components that were not present in the original two-tone input. Depending on the nonlinearity, products can include:

$f_2 - f_1$
 $f_1 + f_2$
 $2f_1 - f_2$
 $2f_2 - f_1$
 $2f_1 + f_2$
 $2f_2 + f_1$

Signal Bench supplies the known stimulus; analysis of the DUT output belongs to Spectral Bench or another analyzer.

For fast, repeatable analyzer testing, Signal Bench includes one-action presets matched to Spectral Bench measurement modes:

- 100 Hz single tone;
- 1 kHz single tone;
- 10 kHz single tone;
- SMPTE-style 60 Hz + 7 kHz, with the 7 kHz component 12.0412 dB below the 60 Hz component, corresponding to a 4:1 amplitude ratio;
- CCIF 19 kHz + 20 kHz, equal amplitudes.

Loading one of these presets selects Dual Sine, establishes the required frequencies and relative tone levels, and disables Low Cut, High Cut, Slicer, added harmonics and Pick Attack shaping. Single-tone presets disable Tone 2.

The preset deliberately preserves **Output Level** and **Mute**. A preset can therefore establish the spectral relationship without unexpectedly changing the user's overall test level or unmuting the generator.



Figure 3.2. Signal Bench measurement presets provide deterministic stimuli matched to common Spectral Bench analyses.

3.8 Guitar-oriented interval presets

Dual Sine can also load two-note stimuli based on guitar-oriented roots from E2 through E5. Available intervals are minor 3rd, major 3rd, 5th and octave.

The preset system supports both 12-tone equal temperament and simple Just ratios.

For 12-TET:

$$f_2 = f_1 * 2^{(n/12)}$$

where n is 3, 4, 7 or 12 semitones for the available intervals.

The Just ratios are:

Interval	Ratio
minor 3rd	6/5
major 3rd	5/4
5th	3/2
octave	2/1

--	--

Loading a guitar interval preset writes the calculated values into the normal F1 and F2 parameters and enables Tone 2. The frequencies remain freely adjustable afterward.

These presets are useful for comparing nonlinear behavior with equal-tempered and pure-ratio input tones. They are measurement conveniences, not guitar-string models.

3.9 Harmonic enrichment

The optional **HARMONICS** function adds H2 through H5 to each Dual Sine tone. It is Off by default.

Harmonic amplitude follows:

$$\text{amplitude}(n) = 1 / n^2$$

relative to that tone's fundamental.

Harmonic	Relative level
H2	-12.04 dB
H3	-19.08 dB
H4	-24.08 dB
H5	-27.96 dB

A harmonic is omitted when its frequency reaches or exceeds $0.48 * \text{sample_rate}$, leaving a guard below Nyquist.

This is a deterministic way to create a spectrally richer known stimulus. It is not intended to model the harmonic structure of a real guitar string.

3.10 Slicer

The Slicer is available in Dual Sine mode and is Off by default. Its rate is adjustable from 0.2 to 5.0 Hz, with a default of 2.0 Hz. Two shapes are available: **Gate** and **Guitar**.

Gate

Gate is a deterministic 50% duty-cycle amplitude gate with 12 ms raised-cosine rise and fall edges. The finite edges reduce discontinuities compared with an instantaneous rectangular gate.

Guitar

Guitar shape provides a deterministic picked-chord-style amplitude envelope with:

- a fixed 5 ms raised-cosine onset;
- 35% fast-decay component;
- 65% slow-decay component;
- fixed fast time constant of 120 ms;

- adjustable slow decay from 0.2 to 5.0 s, default 2.0 s.

Conceptually:

```
body(t) =
    0.35 * exp(-t / 0.120)
    + 0.65 * exp(-t / slow_decay)
```

The beginning of each cycle is made continuous with the mathematically calculated end level of the previous cycle rather than being forcibly reset to zero. The selected Slicer Rate determines the retrigger period.

3.11 Pick Attack

Pick Attack is an optional deterministic excitation available only when Dual Sine, Slicer, Guitar shape and Pick Attack are all enabled. It is Off by default.

The excitation begins with a deterministic xorshift32 sequence. The PRNG and filter states are reset identically on every Slicer retrigger, so each Pick Attack event is repeatable.

The source passes through an adjustable first-order low-pass and a fixed high-pass-style emphasis. **PICK LPF** ranges from 500 Hz to 12 kHz, default 1800 Hz. The low-pass coefficient is sample-rate aware:

$$a = \exp(-2\pi \text{fc} / F_s)$$

The event lasts 5 ms, with a 3 ms raised-cosine attack, an approximately 1.7 ms exponential decay time constant and a final 1.5 ms cosine tail that returns the event smoothly to zero.

PICK LEVEL ranges from -20.0 to 0.0 dB, default -4.0 dB. It is not additionally multiplied by the Tone 1 or Tone 2 level controls. The Pick Attack is, however, multiplied by the same Guitar amplitude envelope as the tonal signal, tying it to the retriggered event rather than adding an unrelated click.

3.12 Signal flow

The normal Dual Sine path is:

```
Tone path: F1 → frequency smoothing → oscillator → optional harmonics → Tone 1 Level
Tone 2 path: F2 → frequency smoothing → oscillator → Tone 2 Level
Tone paths (sum) → Slicer/Guitar envelope → Pick Attack processing → Output Level → Mute → output
```

With Guitar Pick Attack enabled, its deterministic noise/filter/time-envelope path joins the two tonal paths before master Output Level.

Pink and White use the simpler path:

```
Noise path: independent channel PRNG → Pink algorithm (Pink mode) → Low Cut → High Cut
→ Output Level → Mute → output → channel meter / peak capture
```

3.13 Output Level, Mute and metering

Master **Output Level** ranges from -60.0 to 0.0 dB and defaults to -18.0 dB. It follows the bandwidth filters in Pink/White mode and follows all tonal, harmonic, envelope and Pick Attack components in Dual Sine mode.

The 0 dB maximum is intentional. Pink noise has relatively modest RMS level because of its broadband statistical waveform and crest factor, but the validated generator still produces instantaneous peaks close

to full scale. Increasing broadband level through hidden limiting, compression, clipping or peak normalization would alter the stimulus.

MUTE zeros the final output after generation and gain processing. Metering follows that final value, so the meters fall toward their floor while muted.

Signal Bench 2.0.0 presents independent left/right peak meters with engineering dBFS markings at 0, -6, -12, -18, -24, -36, -48 and -60 dBFS. The DSP peak comes from the same final sample written to the output buffer.

The visible meter may react before the corresponding sound emerges from the physical output because the UI sees the generated buffer before downstream host, driver and hardware buffering reaches the DAC. No fixed artificial meter delay is added because the required compensation would depend on the system. GUI-only meter smoothing uses fast attack and approximately 0.5 s release and does not alter the audio.

The meters are useful output indicators, not precision analyzer channels.

3.14 State persistence

Signal Bench preserves its operating state between sessions, including signal type, master level, bandwidth filters, Mute, F1/F2, tone levels, Slicer settings, Guitar decay, Pick Attack settings and Harmonics state.

The Root, Interval and Tuning selectors are convenience controls. Loading a preset calculates and writes the resulting values into the ordinary persistent parameters.

3.15 macOS formats and installation

Signal Bench 2.0.0 is provided on macOS as:

- Standalone;
- Audio Unit;
- VST3.

The Audio Unit is a Music Device, allowing Logic Pro to expose Signal Bench as a Software Instrument.

The current identities are:

Bundle identifier:	works.60n.signalbench
AU identity:	aumu / SgB1 / Sn60
Vendor:	60°N Signal Works

The individual macOS package installs:

```
/Applications/60°N Signal Works Audio Bench Suite/Signal Bench.app  
/Library/Audio/Plug-Ins/Components/Signal Bench.component  
/Library/Audio/Plug-Ins/VST3/Signal Bench.vst3
```

At the documented 2.0.0 release checkpoint, the package is unsigned. macOS may therefore require explicit approval in **System Settings > Privacy & Security** when the package is obtained from a trusted source. Signing and notarization are distribution concerns and do not change the signal-generation algorithms described in this chapter.

3.16 Measurement confidence

Signal Bench has been checked against known signal-generation cases and real macOS audio operation. For the user, the important point is that the generated signals, levels, filtering and metering are intended to be predictable measurement stimuli rather than merely convenient test sounds.

Pink-noise spectral and RMS behavior has been checked at 44.1, 48, 96 and 192 kHz. Standalone, Audio Unit and VST3 operation has also been verified on the qualified macOS release.

These checks establish confidence in the generator under the tested conditions; they do not replace verification of the user's own interface gain, routing, sample rate or DUT state.

3.17 Practical workflows

Generate a known single tone

1. Select the appropriate 100 Hz, 1 kHz or 10 kHz measurement preset.
2. Confirm Output Level and Mute.
3. Verify the physical output level at the DUT when analog calibration matters.
4. Use Spectral Bench or another analyzer to inspect the DUT output.

Run an IMD measurement

Select the SMPTE-style or CCIF preset corresponding to the intended Spectral Bench analysis. The preset establishes the required frequency and amplitude relationship while preserving master Output Level and Mute.

Exercise a nonlinear guitar-oriented device

Load a Root + Interval preset, choose 12-TET or Just tuning as required, and first measure with Harmonics, Slicer and Pick Attack disabled. Add those deterministic components one at a time when the experiment specifically requires a richer or time-varying stimulus. This keeps the cause of any observed change identifiable.

Generate bandwidth-limited noise

Select Pink or White, enable Low Cut and/or High Cut, choose the required slope, and verify that the selected cutoff remains meaningful at the current sample rate. Remember that the Output Level control is a gain control, not a calibrated broadband RMS-output setting.

3.18 Interpretation and cautions

Signal Bench is deliberately transparent about what it does not do:

- it does not calibrate dBFS to analog volts;
- it does not automatically prevent clipping with a limiter;
- it does not calculate the distortion products produced by a DUT;
- its Guitar envelope and Pick Attack are deterministic measurement stimuli, not physical instrument models;
- its output meters are indicators, not calibrated analysis channels;
- a low broadband RMS value does not imply unused peak headroom.

For measurements that depend on exact spectral interpretation, use Signal Bench as the known source and let Spectral Bench perform the calibrated analysis.

4. Spectral Bench

Spectral Bench 2.1.0 for macOS is the suite's real-time spectrum and audio-measurement analyzer. Its spectrum display is the primary instrument, with calibrated level, tone, harmonic, distortion, intermodulation, sweep and phase measurements layered on the same analysis foundation.

The application is designed around explicit measurement definitions. Display zoom and trace presentation are kept separate from authoritative measurement math, and measurements that require a reference, bandwidth or phase convention expose that convention rather than hiding it.

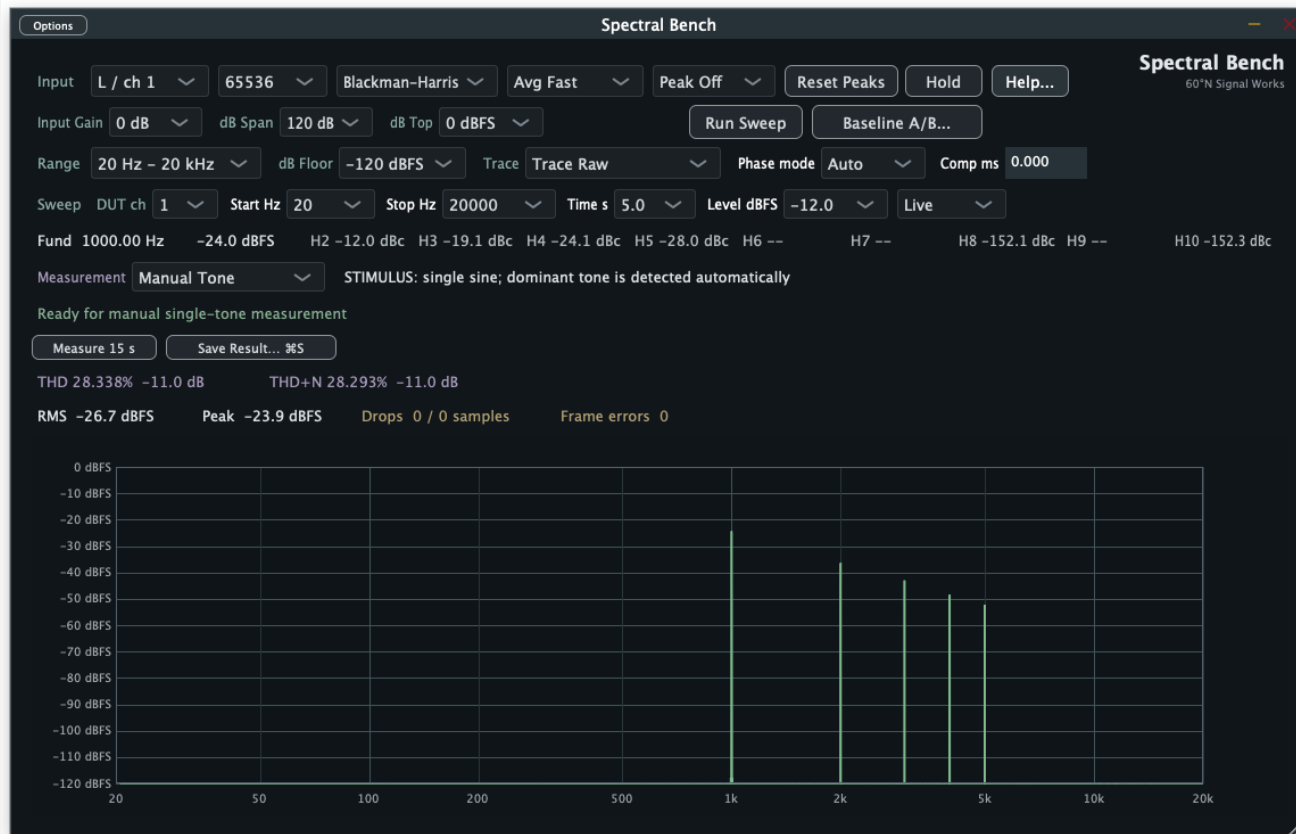


Figure 4.1. Spectral Bench 2.1.0 configured for live spectrum inspection.

4.1 General setup

Select the input channel that actually carries the signal under test. Start with **Input Gain** = 0 dB and keep enough headroom that neither the DUT nor the interface clips. For generated sweeps, -12 dBFS is a useful general starting level.

The normal overview is 20 Hz to 20 kHz, but the viewport can be narrowed substantially without changing the underlying calibrated measurement. Frequency view and dB Top/Floor/Span are display controls. **Input Gain is different:** it changes the samples presented to the analyzer and therefore changes measured dBFS values.

A practical live-spectrum starting point is FFT 16384, Blackman-Harris 4-term, Averaging Off for immediate response or Medium for a steadier trace, Peak Off, Raw trace and Input Gain 0 dB.

Save Result... (§S) writes PNG, CSV and TXT companions. The PNG preserves the visual state, while CSV/TXT preserve the numeric and provenance information.

4.2 Spectrum analyzer

The real-time analyzer provides:

- logarithmic frequency display;
- calibrated dBFS magnitude;
- FFT sizes 4096, 8192, 16384, 32768 and 65536;
- Hann, 4-term Blackman-Harris and Flat Top windows;
- Off/Fast/Medium/Slow averaging;
- Off/Hold/Decay peak behavior with explicit reset;
- Raw and Noise Smooth trace presentation;
- adjustable frequency and dB viewports;
- a free crosshair with fixed Trace/Cursor readouts.

The default FFT size is 16384.

4.2.1 Frequency viewport

macOS 2.1.0 includes logarithmic presets covering useful low-, mid- and high-frequency regions, plus a Custom minimum/maximum view with adaptive 1-2-5 engineering grid generation. The frequency axis remains logarithmic.

Changing the visible frequency range does not change analyzer calibration or measurement bandwidth. It changes what part of the spectrum is shown.

4.2.2 dB viewport and Input Gain

The dB viewport can range from a 120 dB overview down to a 10 dB span. dB Top, Floor and Span are coupled: selecting Top or Floor establishes the zoom anchor, and later Span changes preserve that boundary while moving the other.

Input Gain, adjustable from -20 to +40 dB, is not a display control. It modifies the samples entering the analyzer. Use it deliberately and record it when absolute dBFS results matter.

4.2.3 Cursor and trace readout

The crosshair reports mouse frequency and dBFS position, the trace level at the same frequency and their signed dB difference. The readout remains in a fixed top-right location so it does not chase the pointer.

When a result is saved, the active cursor/trace measurement is included in the PNG, CSV and TXT report.

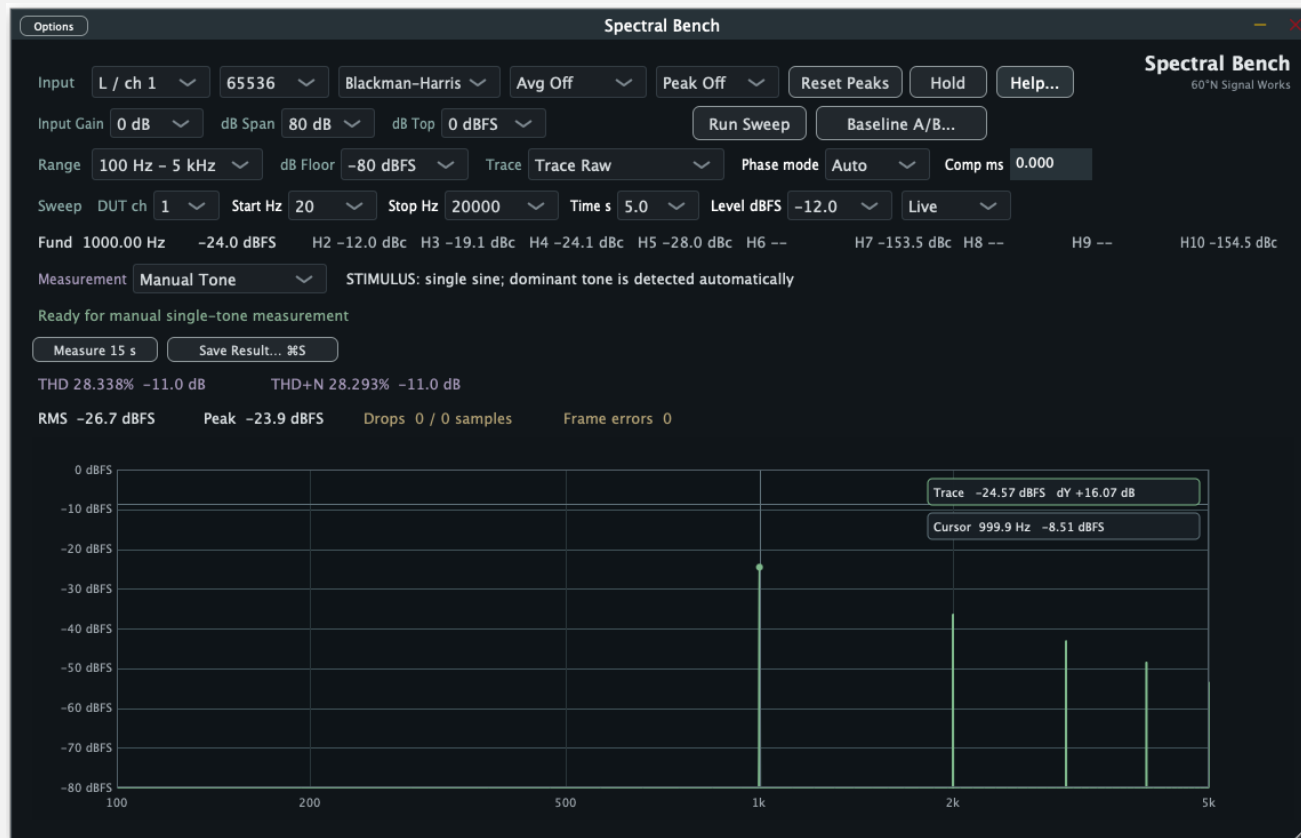


Figure 4.2. Viewport zoom and cursor measurement in Spectral Bench.

4.3 FFT calibration and level semantics

Spectral Bench distinguishes time-domain level from spectral tone amplitude.

Sample Peak is the maximum absolute input sample over the relevant analysis interval. **RMS** is calculated from the time-domain samples. Neither depends on FFT size, FFT window, overlap or averaging.

Spectral tone amplitude comes from the calibrated FFT/fit path. Window coherent gain is explicitly accounted for so that a bin-centered coherent sine reports its intended amplitude. Window equivalent noise bandwidth (ENBW) is also an explicit property because broadband noise and a discrete tone cannot be interpreted with the same per-bin assumptions.

The application does not claim a dBFS/Hz power-spectral-density representation. Broadband noise therefore needs to be interpreted using the analyzer's documented spectrum semantics rather than as if each displayed value were automatically a density measurement.

4.4 FFT windows

The supported windows are Hann, 4-term Blackman-Harris and Flat Top.

Use Blackman-Harris as a strong general-purpose choice when spectral leakage rejection matters. Flat Top is useful when amplitude accuracy for isolated tones is more important than narrow main-lobe width. Hann is a conventional compromise with a relatively narrow lobe and moderate leakage suppression.

The important engineering point is that changing the window changes leakage, main-lobe width, coherent gain and ENBW. Spectral Bench calibrates the supported windows, but no window removes the underlying time/frequency-resolution tradeoff.

4.5 Averaging, Peak Hold and trace presentation

Averaging operates in **linear power** and is converted to dB afterward:

$$\begin{aligned} \text{Pavg_new} &= \alpha * \text{Pavg_old} + (1 - \alpha) * \text{Pnew} \\ \alpha &= \exp(-\text{hopSeconds} / \tau) \end{aligned}$$

The implemented time constants are:

Mode	Time constant
Fast	0.25 s
Medium	1.00 s
Slow	4.00 s

Peak behavior is Off, Hold or Decay. Decay uses 20 dB/s, and Reset clears stored peak state.

Noise Smooth is display-only. It is useful for visually stabilizing broadband Pink or White noise, especially at high frequencies. It does not modify the calibrated measurement snapshot, cursor source data, harmonic analysis, THD, THD+N, IM products or authoritative numeric export.

Hold, which freezes the displayed graph, is also distinct from Peak Hold/Decay.

4.6 Single-tone measurements

Spectral Bench provides defined 100 Hz, 1 kHz and 10 kHz modes plus Manual Tone.

For an applicable single-tone measurement, the panel can report:

- input RMS and sample peak;
- nominal frequency where defined;
- measured fundamental frequency;
- fitted fundamental amplitude;
- H2 through H10 where valid below Nyquist;
- each harmonic relative to the fundamental in dBc;
- THD;
- THD+N;
- actual THD+N bandwidth.

Defined-frequency modes search around the expected tone rather than simply choosing any dominant spectral component. Manual Tone performs dominant-tone detection across the normal analysis range. A tone-confidence requirement prevents an arbitrary weak spectral maximum from being promoted to a valid measurement.

Signal Bench provides matching deterministic 100 Hz, 1 kHz and 10 kHz stimulus presets.

4.7 Harmonics and THD

Harmonic analysis fits the fundamental and valid harmonics H2 through H10. Harmonic levels are reported in dBc relative to the fitted fundamental.

THD is conceptually:

$$\text{THD} = \sqrt{V_2^2 + V_3^2 + \dots + V_n^2} / V_1$$

where only valid harmonics within the supported analysis range contribute.

THD describes harmonic products only. It does not include unrelated broadband noise and must not be used interchangeably with THD+N.

4.8 THD+N

THD+N removes the fitted fundamental and integrates the remaining residual power over an explicit measurement bandwidth. The measurement therefore includes harmonic distortion, noise and any other residual energy inside that bandwidth.

The available bandwidth semantics include 20 Hz to 20 kHz where the sample rate permits it and 20 Hz to Nyquist. No A-weighting, AES weighting or other weighting filter is part of the documented measurement.

Always report the THD+N bandwidth with the result. A percentage without its measurement bandwidth is incomplete.

4.9 Fifteen-second single-tone statistics

Manual Tone and the three defined single-tone modes can accumulate a 15-second measurement window using new accepted analysis frames. SMPTE-style and CCIF modes do not use this helper because they report selective products rather than a normative aggregate measurement.

The statistics function is useful when a single live value is too volatile to describe a DUT. It does not turn an unstable or invalid stimulus into a valid one; the same stimulus/confidence rules still apply to accepted frames.

4.10 CCIF / DFD-style 19 + 20 kHz measurement

The defined stimulus is:

f1 = 19 kHz
f2 = 20 kHz
equal amplitudes

The difference frequency is 1 kHz. Spectral Bench explicitly inspects:

f2 - f1	= 1 kHz
2*f1 - f2	= 18 kHz
2*f2 - f1	= 21 kHz

The 21 kHz product is valid only when it is below Nyquist and inside the usable analyzer range.

Numerical analysis may report it even though the normal live spectrum view is capped at 20 kHz; display range and analysis range are different concepts.

Available products are reported in dBc.

The wording **CCIF / DFD-style** is intentional. Spectral Bench does not claim a normative aggregate CCIF/ITU-R percentage because the exact standard normalization has not been established from the normative source.

4.11 SMPTE-style 60 Hz + 7 kHz measurement

The defined stimulus is:

fL = 60 Hz

fH = 7 kHz

low/high amplitude ratio = 4:1

A 4:1 amplitude ratio is approximately 12.041 dB.

The analyzer explicitly inspects the second-order sideband pair at 6940 and 7060 Hz and the third-order pair at 6880 and 7120 Hz. Products are reported numerically in dBc.

Again, **SMPTE-style** is deliberate terminology. The selective sideband measurements are fully defined, but Spectral Bench does not present them as a normative aggregate SMPTE IMD percentage.

Signal Bench includes matching SMPTE-style and CCIF presets, reducing the chance that generator and analyzer settings are accidentally mismatched.

4.12 Referenced frequency-response sweep

macOS 2.1.0 adds an automated logarithmic sine sweep using a fixed two-channel referenced topology. Select **DUT ch 1** or **DUT ch 2**: that channel is the DUT output/input path and the other channel automatically becomes the direct reference path. Both outputs receive the same sample-synchronous deterministic sweep. There is no separate reference-channel selector.

For a referenced transfer measurement, loop the reference channel directly back and route the selected DUT channel through the device under test. Transfer magnitude is calculated as DUT/Reference, so response common to both paths cancels.

A direct dual-channel loopback is the best first reference check. It exposes physical channel/path mismatch and confirms the two-channel topology before a DUT is introduced. Swapping DUT channel swaps both output and input roles.

Use **Transfer** when the question is the DUT response relative to the simultaneous reference. Use **Actual** when the absolute captured DUT response is the quantity of interest.

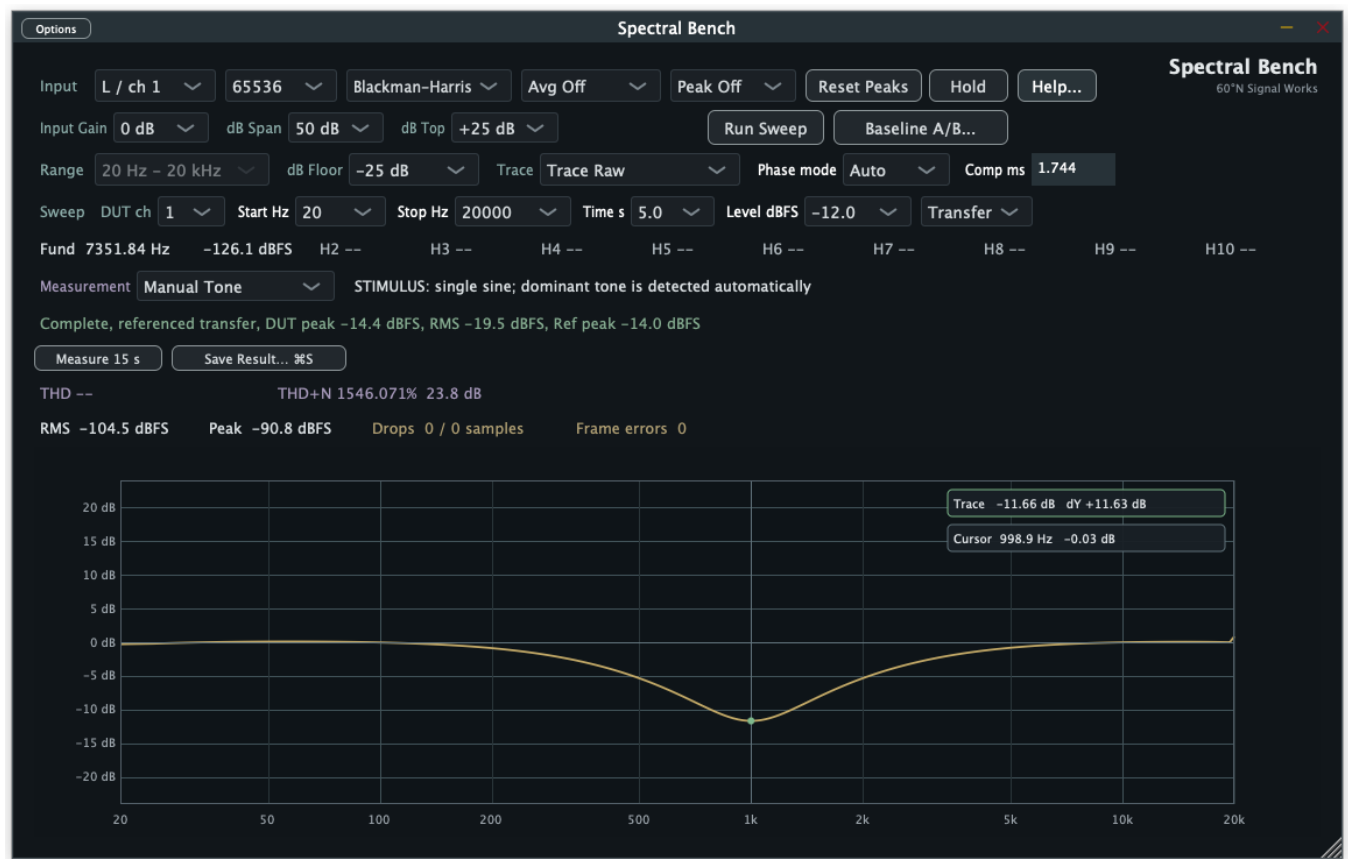


Figure 4.3. Referenced DUT transfer sweep.

4.13 Transfer versus Actual response

Transfer answers the relative question: what did the DUT path do compared with the simultaneously measured reference path?

Actual shows the captured DUT response itself. It is useful when an absolute system response is wanted or when a suitable reference path is not available.

Do not interpret an unreferenced Actual trace as if interface, cable and channel response had automatically been divided out. Conversely, a referenced Transfer measurement is only as meaningful as the reference path and channel matching used to create it.

4.14 Phase modes

Phase is inherently reference-dependent. Spectral Bench 2.1.0 therefore exposes four distinct modes rather than pretending there is one universally correct phase trace.

Raw

Raw preserves the measured phase relationship, including transport delay. A constant delay appears as a frequency-dependent phase slope and repeated wraps.

Use Raw when that complete measured relationship is the quantity of interest.

Auto

Auto estimates and removes a response-dependent constant-delay component to make the remaining phase easier to inspect. This is a presentation/reference choice, not an independent latency measurement.

The delay/linear-phase ambiguity means that a frequency-response measurement alone cannot always separate pure transport delay from linear phase belonging to the DUT response. Auto can therefore produce a useful phase view without equalling the physical latency measured by Latency Bench.

Manual

Manual applies a user-specified delay compensation. This is the correct mode when the transport latency is known independently and the goal is to inspect phase after removing that known constant-delay component.

For example, a physical path measured by Latency Bench at 48 kHz produced 89.15 samples / 1.857 ms. Entering that independently measured value in Spectral Bench Manual phase removed the expected constant-delay component and produced an essentially flat residual phase.

Baseline A/B

Baseline uses a stored reference measurement and shows subsequent phase relative to it. The guided A/B workflow is particularly useful for measurements such as filter Off versus filter On, where the common transport delay should disappear and the phase change caused by the DUT state is the desired result.

In a reference check, LPF Off as baseline followed by a 1 kHz LPF On measurement produced the expected filter phase without transport-delay wraps. Off followed by Off produced an essentially 0-degree null across the displayed band.

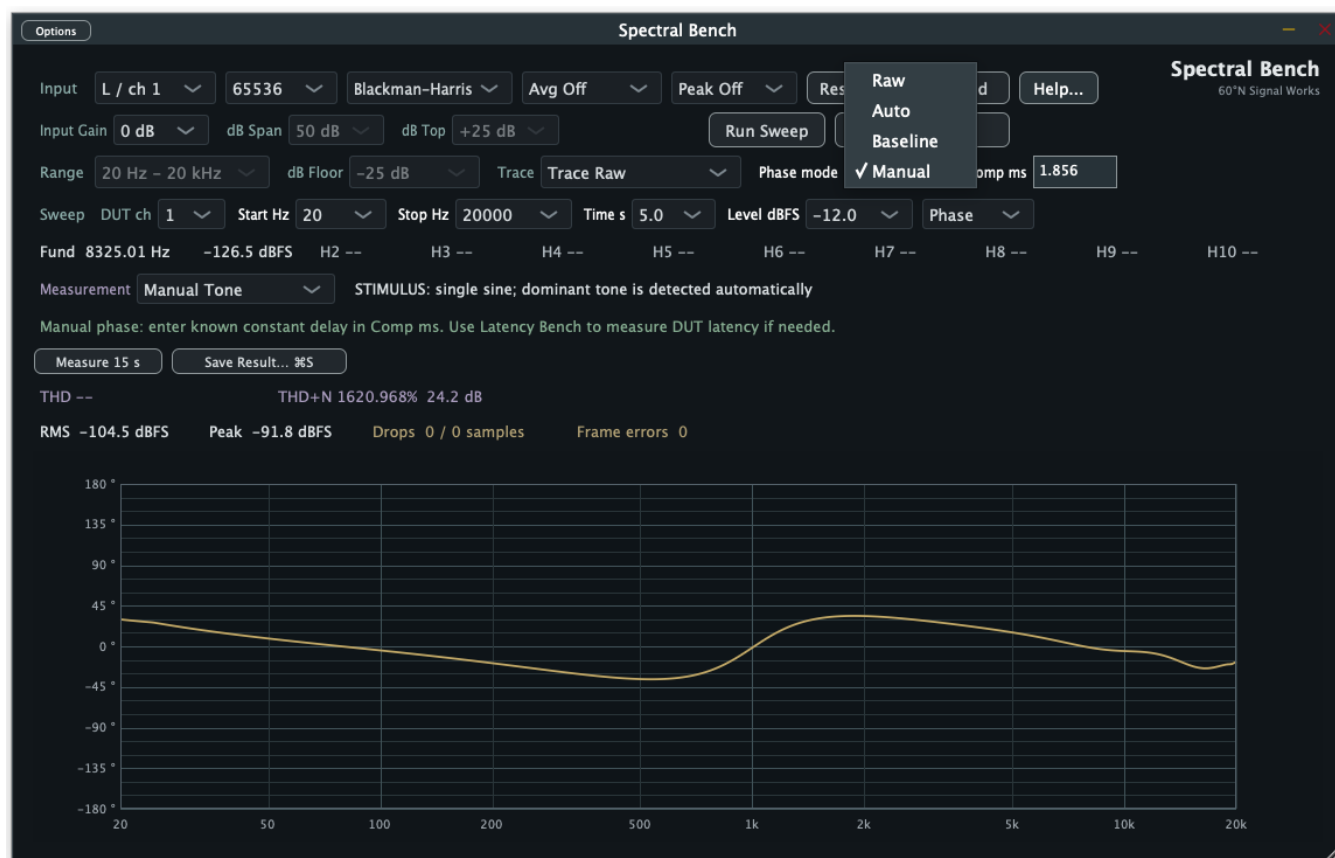


Figure 4.4. Spectral Bench phase-reference modes.

4.15 Why phase and latency must not be confused

A measured transfer function cannot in general distinguish an arbitrary constant transport delay from an equivalent linear phase term solely from the frequency-domain response. This is why Spectral Bench Auto compensation and Latency Bench can legitimately report different delay values for the same physical DUT path.

In one physical example with a Neural DSP Quad Cortex, Latency Bench measured 89.15 samples / 1.857 ms at 48 kHz while Spectral Bench Auto estimated about 1.42 ms for the tested response. Manual compensation with the independently measured 1.857 ms behaved as expected because the two tools were answering different timing questions.

Use Latency Bench when the quantity required is physical-path latency. Use Spectral Bench phase modes when the quantity required is phase referenced according to Raw, Auto, known-delay Manual or A/B Baseline semantics.

4.16 Saving a result

Save Result... (⌘S) creates three companion files:

- PNG, preserving the visual graph and active cursor/trace annotation;
- CSV, for structured numeric data;
- TXT, for a human-readable report and provenance.

The export terminology distinguishes live analyzer state from completed sweep state. CSV uses `analyzer_mode` for the analyzer selector and `sweep_type` for the appended sweep. TXT uses Analyzer mode: and Sweep type:. Phase exports record the selected phase mode and, for Auto or Manual, the applied compensation in milliseconds.

This distinction matters when a report contains both a live analyzer mode such as Manual Tone and a completed referenced sweep.

4.17 macOS formats and installation

Spectral Bench 2.1.0 for macOS is available as Standalone, Audio Unit and VST3. Use the format that best fits the measurement workflow; the measurement definitions in this chapter are the same across those macOS formats.

The handbook focuses on the qualified macOS release. Older cross-platform release history belongs in project/release documentation rather than the operating flow of this chapter.

4.18 Measurement confidence

Spectral Bench has been checked with known FFT/window, level, tone, harmonic, THD/THD+N, intermodulation, sweep and phase cases, together with physical loopback and representative DUT measurements.

Physical checks included 44.1, 48 and 96 kHz and representative buffer sizes. Direct loopback and channel-swap measurements were used to distinguish analyzer behavior from real channel/path mismatch.

For a user measurement, the practical consequence is simple: establish a clean reference path first, then treat the DUT result as valid only while the same sample rate, channels, levels and reference conditions remain in force.

4.19 Practical measurement workflows

Live spectrum or noise inspection

Use Raw trace for detailed spectral structure. Medium averaging is a useful starting point for a steadier broadband display; Noise Smooth can make a Pink/White noise trace visually easier to read without changing numeric measurements. Use cursor/trace delta for local inspection.

Single-tone distortion

Use a matching Signal Bench 100 Hz, 1 kHz or 10 kHz preset when possible. Confirm a valid fundamental, then read H2-H10, THD and THD+N. Record the THD+N bandwidth.

Intermodulation products

Use the matching Signal Bench SMPTE-style or CCIF preset. Interpret the explicitly reported sidebands/products in dBc. Do not relabel them as normative aggregate SMPTE or CCIF percentages.

DUT frequency response

First run a direct referenced loopback to qualify the two channels. Then insert the DUT in the DUT path and run Transfer. Use Actual only when the absolute captured response is what you intend to report.

Filter phase

If transport delay should remain part of the result, use Raw. If the transport latency is known independently, use Manual and enter that value. For an A/B change such as filter Off versus On, Baseline is usually the clearest reference. Treat Auto as a useful response-dependent compensation, not as a substitute for Latency Bench.

4.20 Common interpretation errors

The most important traps are conceptual rather than numerical.

A narrower display range does not create a narrower measurement bandwidth. Input Gain is not merely a visual zoom. THD and THD+N are not interchangeable. A selective IM product is not automatically a normative aggregate IMD percentage. An Actual response is not automatically a referenced transfer function. Auto phase compensation is not a latency measurement. And a very stable result can still be wrong if the reference channel, DUT path or physical wiring is wrong.

When the result matters, save it together with enough setup information to reconstruct what was actually measured.

5. Latency Bench

Latency Bench 1.1.1 for macOS measures **physical-path latency** through an audio device or DUT by comparing a direct reference path with a second path through the DUT. It is designed for measurements where cable, converter, analog, digital and processing delays are part of the real signal path.

The application generates its own deterministic broadband probe, captures the reference and DUT responses, estimates their relative timing by normalized cross-correlation and reports the DUT path relative to a stored baseline.

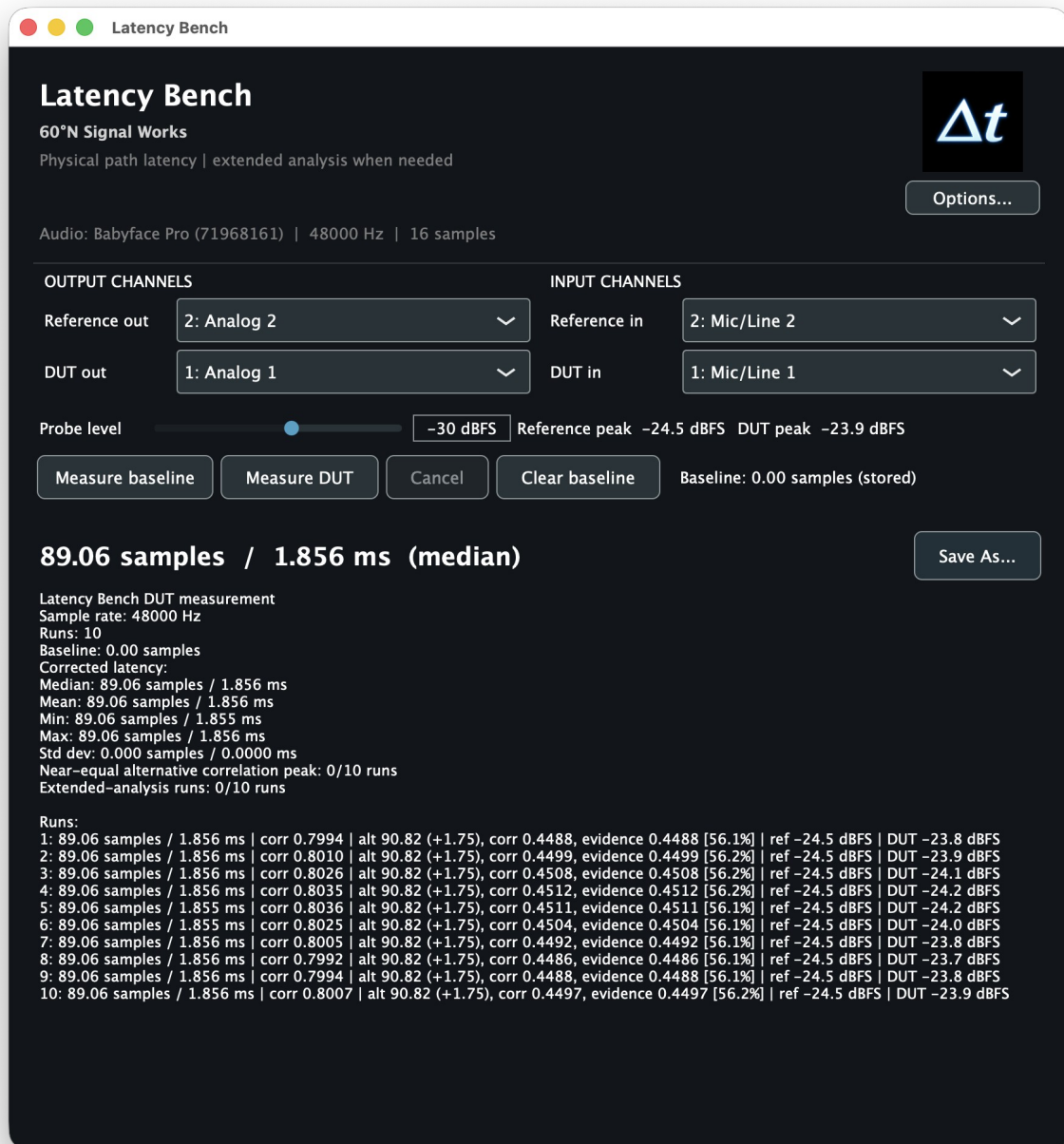


Figure 5.1. Latency Bench 1.1.1 configured for a physical-path DUT measurement.

5.1 What Latency Bench measures

Latency Bench measures the timing difference between two simultaneously defined physical paths:

Reference OUT A -> direct reference path -> Reference IN A

DUT OUT B -> device under test -> DUT IN B

The baseline characterizes the difference between the two measurement loops without the DUT contribution. A later DUT measurement subtracts that baseline from the measured reference-to-DUT offset.

This makes the result a physical-path measurement rather than a calculation from driver buffer settings.

Depending on the DUT, the reported timing can include:

- interface output and input conversion;
- analog circuitry;
- DSP transport;
- internal buffering;
- effects-loop paths;
- sample-rate conversion;
- filter phase/group delay;
- any other delay that changes the captured response relative to the reference.

That last point is important. Latency Bench measures the timing of the **captured response**. With a strongly bandwidth-limited or phase-shifting DUT, that is not necessarily identical to bare digital transport latency.

5.2 Wiring

For the baseline, make both paths as equivalent as practical:

OUT A -> cable -> IN A

OUT B -> cable -> IN B

For the DUT measurement, leave the reference path intact and insert the DUT into path B:

OUT A -> cable -> IN A

OUT B -> DUT -> IN B

Use the same interface, sample rate, channel assignments and physical reference path for baseline and DUT measurement.

Latency Bench physical wiring

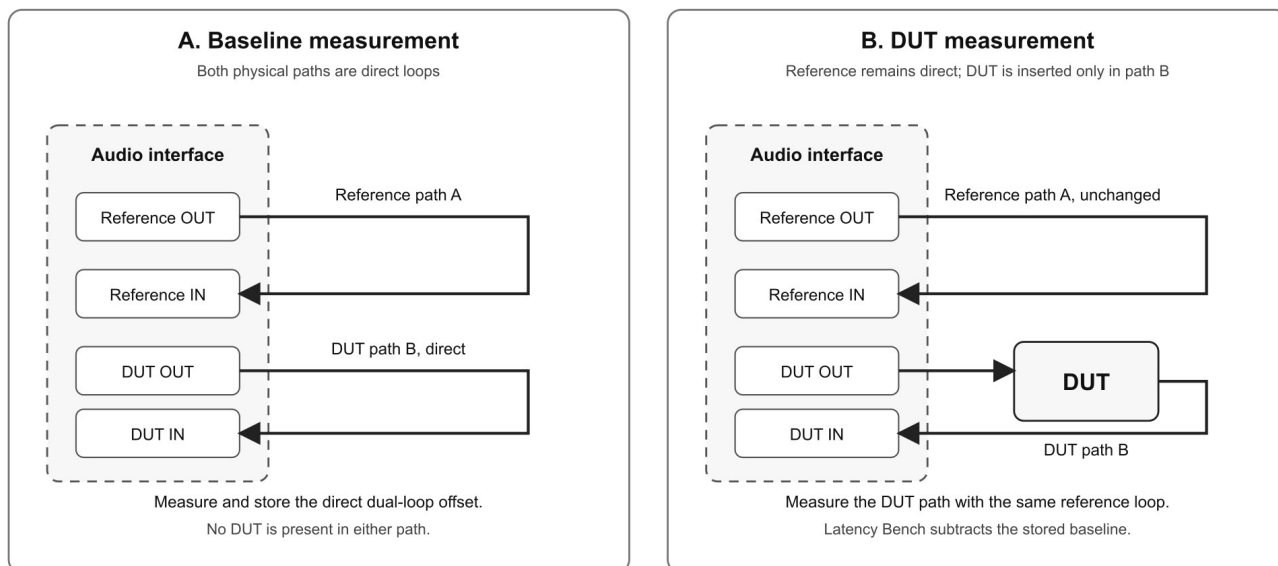


Figure 5.2. Baseline and DUT physical-path wiring.

5.3 Measurement workflow

A reliable workflow is:

1. Select the audio device and the Reference OUT/IN and DUT OUT/IN channels.
2. Set a conservative probe level. The default is -30 dBFS.
3. Wire both paths directly and run **Measure Baseline**.
4. Confirm that the baseline is accepted and stable.
5. Insert the DUT into path B without changing the reference path.
6. Put the DUT into the exact state to be measured.
7. Run the DUT measurement.
8. Interpret the reported median, run spread, correlation/evidence and any competing timing candidate.
9. Save or record the result together with the physical setup and DUT state.

The DUT measurement performs ten automatic runs. The **median** is the primary reported result because it is robust against an occasional outlying run. Mean, minimum, maximum and standard deviation provide additional repeatability evidence.

5.4 Probe signal

Latency Bench uses a deterministic pseudo-random bipolar broadband probe. The same known sequence is sent through the reference and DUT output paths.

The normal analysis configuration uses:

probe length:	2048 samples
capture length:	32768 samples
prime:	2048 samples
tail:	12288 samples
maximum lag:	+/-16384 samples
correlation window:	4096 samples

The deterministic probe gives the estimator a known broadband signature rather than relying on an arbitrary program signal or a single periodic tone.

A periodic sine alone is a poor general-purpose latency probe because many delays separated by an integer number of periods can appear equally plausible. Broadband structure greatly reduces that ambiguity for ordinary paths.

5.5 Normalized cross-correlation

For each candidate lag, the estimator compares overlapping reference and DUT samples using normalized cross-correlation. Conceptually:

$$r(k) = \sum_n x[n] y[n+k] / \sqrt{(\sum_n x[n]^2)(\sum_n y[n+k]^2)}$$

Normalization makes the timing comparison substantially insensitive to simple gain differences.

Latency Bench searches both positive and negative lags. It uses the **magnitude** of correlation for timing selection, so a polarity-inverted but otherwise valid path can still be timed correctly. The sign of raw correlation remains useful evidence about the relationship between the captured signals.

The correlation peak is refined with parabolic interpolation, allowing fractional-sample timing estimates.

5.6 Correlation evidence and overlap weighting

A raw normalized-correlation value can become misleading near the edges of a search, where only a small number of samples overlap. Version 1.1.1 therefore keeps raw correlation separate from the evidence used to select the winning lag.

The selection evidence is:

$$E = |r| \sqrt{(N_o / N_w)}$$

This penalizes candidates supported by unusually short overlap without adding a preferred latency direction, expected-latency bias or polarity bias.

Here $|r|$ is the normalized-correlation magnitude, N_o is the number of overlapping samples for the candidate lag, and N_w is the nominal analysis-window length.

5.7 Confidence and ambiguity rejection

Latency Bench is intentionally allowed to say that a measurement is ambiguous.

The primary timing candidate must have evidence of at least 0.10. After selecting it, the estimator searches for a sufficiently separate competing candidate. If the strongest separate candidate reaches at least **90% of the primary evidence**, the measurement is rejected rather than reporting an arbitrary winner.

This is especially important for narrow-band or strongly resonant responses, where the correlation function can contain several plausible lobes.

A rejected measurement is useful information. It means the captured response did not provide enough unique timing evidence under the current conditions to support one interpretation confidently.

5.8 Automatic extended analysis

Normal analysis is deliberately lightweight. When a DUT run has weak or competing timing evidence, version 1.1.1 automatically retries that run with extended analysis.

The extended configuration uses:

probe length: 8192 samples

correlation window: 12288 samples

The longer probe provides more information for strongly filtered paths.

The UI identifies this behavior as:

Physical path latency | extended analysis when needed

For an ordinary broadband DUT, normal analysis is normally sufficient; extended analysis is reserved for cases where the captured evidence requires it.

5.9 Magnitude-matched extended probe

A strongly filtered DUT may remove so much of the broadband probe that the raw reference and DUT no longer resemble each other sufficiently for unambiguous correlation.

During extended analysis, Latency Bench therefore estimates the DUT's captured magnitude spectrum and applies that magnitude shape to the reference probe in the frequency domain before correlation.

The important constraint is that **DUT phase is not copied or corrected**. Only spectral magnitude is matched. The DUT's timing and phase behavior remain in the captured DUT response and are therefore still available to the estimator.

This approach improves timing evidence for legitimate narrow-band and subwoofer-style outputs without defining an arbitrary minimum bandwidth.

The signed lag range, confidence threshold and 90% competing-candidate ambiguity rule remain unchanged.

5.10 No arbitrary bandwidth limit

Latency Bench does not impose a rule such as "the DUT must pass at least N octaves" or "the response must extend above frequency X."

Strongly bandwidth-limited low-frequency paths can still produce valid measurements when the response contains sufficient timing evidence.

Whether a particular path is measurable depends on the timing information present in the captured response, not on a fixed published bandwidth threshold.

The opposite is equally important: some wider-looking responses can still be ambiguous if their correlation structure produces competing candidates. The estimator decides from evidence rather than from a hard-coded audio-band label.

5.11 Polarity reversal

Because timing selection uses correlation magnitude, a simple polarity inversion does not inherently invalidate a latency measurement. An otherwise unchanged inverted response produces a correlation peak of opposite sign at the same timing.

This does **not** mean that arbitrary phase manipulation is irrelevant. Frequency-dependent phase shift, filtering and resonant behavior can alter the waveform and the shape of the correlation function. Polarity inversion is the special case of multiplying the whole waveform by -1.

5.12 Baseline semantics

The baseline is setup-bound. It represents the direct timing difference between the selected physical loops under the current measurement configuration.

If the audio device, sample rate, channel assignments or physical cabling changes, establish a new baseline. Reusing an old baseline after changing the setup can produce a very repeatable but systematically wrong DUT result.

The baseline subtraction can be written conceptually as:

$$\begin{aligned} \text{DUT latency} = & \\ & \text{measured DUT-vs-reference offset} \\ & - \text{stored direct-loop baseline offset} \end{aligned}$$

The baseline is not intended to remove the DUT's own filter phase, converter delay or internal transport. Those are part of the DUT path being measured.

5.13 Complete-response timing versus transport latency

A filtered DUT deserves special care in interpretation.

Suppose a bypass path measures approximately 1.86 ms. A steep low-pass or band-pass path through the same device may measure many milliseconds later even if the device's basic DSP transport has not changed. The additional timing can arise from the phase/group-delay behavior of the filter itself.

Latency Bench therefore reports **complete-response timing** for the captured path.

If the engineering question is specifically "what is the device's bare transport latency independent of this filter?", measure an appropriate broadband/bypass state or use another method capable of isolating that quantity.

If the engineering question is "when does this filtered output actually arrive relative to the reference?", the complete-response result is exactly the relevant quantity.

5.14 Reference physical measurements

A representative physical loopback/bypass reference at 48 kHz measured:

89.15 samples
1.857 ms
standard deviation: 0.000 ms
analysis: normal, 10/10 runs

This provides the comparison point for the filtered-path examples below.

5.14.1 Representative filtered paths

DUT condition	Result	Interpretation
bypass/cable	89.15 samples / 1.857 ms	normal analysis
LPF qualification case	110.71 / 2.306 ms	accepted
300-3400 Hz, 24 dB/oct	95.17 / 1.983 ms	accepted
1 kHz HPF, 48 dB/oct	133.60 / 2.783 ms	accepted
2 kHz HPF, 48 dB/oct	90.91 / 1.894 ms	accepted
2-4 kHz, 48+48 dB/oct	114.05 / 2.376 ms	accepted
20-160 Hz, 24 dB/oct	383.20 / 7.983 ms	extended, accepted
20-160 Hz, 48+48 dB/oct	493.18 / 10.275 ms	extended, accepted

The 20-160 Hz, 48 dB/oct case had approximately 0.006 ms run-to-run standard deviation and used extended analysis for all ten runs. The 24 dB/oct version had approximately 0.005 ms standard deviation and likewise used extended analysis for all ten runs.

Relative to the bypass reference, the 20-160 Hz 48 dB/oct response arrives about 404.03 samples / 8.417 ms later, while the 24 dB/oct response arrives about 294.05 samples / 6.126 ms later. These differences are consistent with measuring the complete filtered response, including filter phase/group delay, rather than merely a fixed DSP transport delay.

5.15 Examples of correctly rejected measurements

Some responses legitimately contain more than one plausible timing candidate; in those cases the measurement can be rejected as ambiguous.

A 2-3 kHz narrow-band case was rejected with a primary candidate around 127.59 samples and a competing candidate around 118.87 samples whose evidence was about 91.6% of the primary.

A 100-500 Hz case was likewise rejected when the competing candidate reached about 92.4% of the primary.

These are successful estimator outcomes. The application refused to convert ambiguous evidence into false precision.

Before magnitude-matched extended analysis was added, a 20-160 Hz 48 dB/oct case could also select a spurious large negative edge candidate. The 1.1.1 overlap-weighted evidence and magnitude-matched extended path were introduced specifically to make such edge/strong-filter cases robust without adding an expected-positive-latency bias.

5.16 Reading the result

A strong result combines several kinds of evidence:

- median DUT latency;
- low run-to-run standard deviation;
- sensible minimum and maximum;
- adequate primary correlation/evidence;
- no competing candidate near the 90% rejection threshold;
- a physical result consistent with the DUT state and measurement definition.

Do not use correlation as a generic audio-quality score. A lower correlation can be perfectly legitimate when a DUT intentionally changes spectral magnitude or phase. Its role here is timing evidence.

Likewise, a standard deviation of zero does not prove absolute accuracy. It proves excellent repeatability under the tested conditions.

5.17 Troubleshooting an ambiguous or weak measurement

If a measurement is rejected or has weak evidence:

1. verify that Reference and DUT channels are not swapped;
2. verify signal level and make sure neither path clips;
3. confirm that the DUT is actually passing the intended signal;
4. check for feedback, parallel paths or unexpected dry signal;
5. repeat the baseline if the physical setup has changed;
6. allow the automatic extended analysis to complete;
7. inspect whether the DUT response is extremely narrow-band or resonant;
8. consider whether the desired quantity is actually complete-response timing or bare transport latency.

Do not solve ambiguity by simply choosing the numerically convenient peak. The rejection rule exists to prevent exactly that.

5.18 Practical examples

Effects-loop latency

Measure the device in a bypass state, then compare the relevant effects-loop configurations while leaving the measurement interface and reference path unchanged. This can reveal latency contributed by analog send/return conversion and routing even when the inserted loop processing itself is bypassed.

Modeler or digital processor

Measure a broadband/bypass state first to establish the device's practical transport/reference timing. Then enable individual DSP blocks or signal paths and compare the resulting complete-response timing. For filters, remember that the difference can include filter group delay rather than only added block scheduling or transport.

Subwoofer or crossover output

A subwoofer output is a legitimate Latency Bench target. Let automatic extended analysis handle the strongly low-pass response. If the estimator accepts the result with good repeatability and no near-equal competitor, there is no requirement to artificially widen the DUT bandwidth.

Interpret the result as the arrival timing of that filtered output.

5.19 Validation scope

Latency Bench has been checked with known delays, polarity inversion, fractional-sample timing, broadband and strongly filtered paths, and deliberately ambiguous cases. Physical loopback and DUT measurements complement the known-answer checks.

The estimator is intentionally allowed to reject a measurement when the captured response does not support one sufficiently dominant timing interpretation. Automatic extended analysis is part of normal operation for difficult bandwidth-limited paths.

Reference measurements in this handbook demonstrate behavior on the tested setups; they are not universal latency specifications for the devices involved.

5.20 Measurement record

For a result worth publishing or comparing later, record:

- Latency Bench version
- audio interface/device
- sample rate
- Reference OUT / IN
- DUT OUT / IN
- probe level
- baseline result
- DUT identity
- DUT state / enabled processing
- median latency in samples and ms
- mean / min / max / standard deviation
- normal or extended-analysis run count

correlation/evidence and competing candidate if relevant
physical wiring notes

A latency number without its signal-path definition is rarely enough to reproduce the measurement.

6. Matrix Bench

Matrix Bench 2.1.1 for macOS is the suite's low-latency routing, mixing and monitoring environment. It combines physical audio devices, an 8-in/8-out virtual Core Audio device, per-input processing, a crosspoint matrix, independent Main and Aux destinations, snapshots, MIDI control and an always-running headless engine.

Unlike a conventional DAW mixer, Matrix Bench is organized around explicit source-to-destination crosspoints. Main and Aux are first-class destinations in the same matrix rather than fixed copies of a single mix.

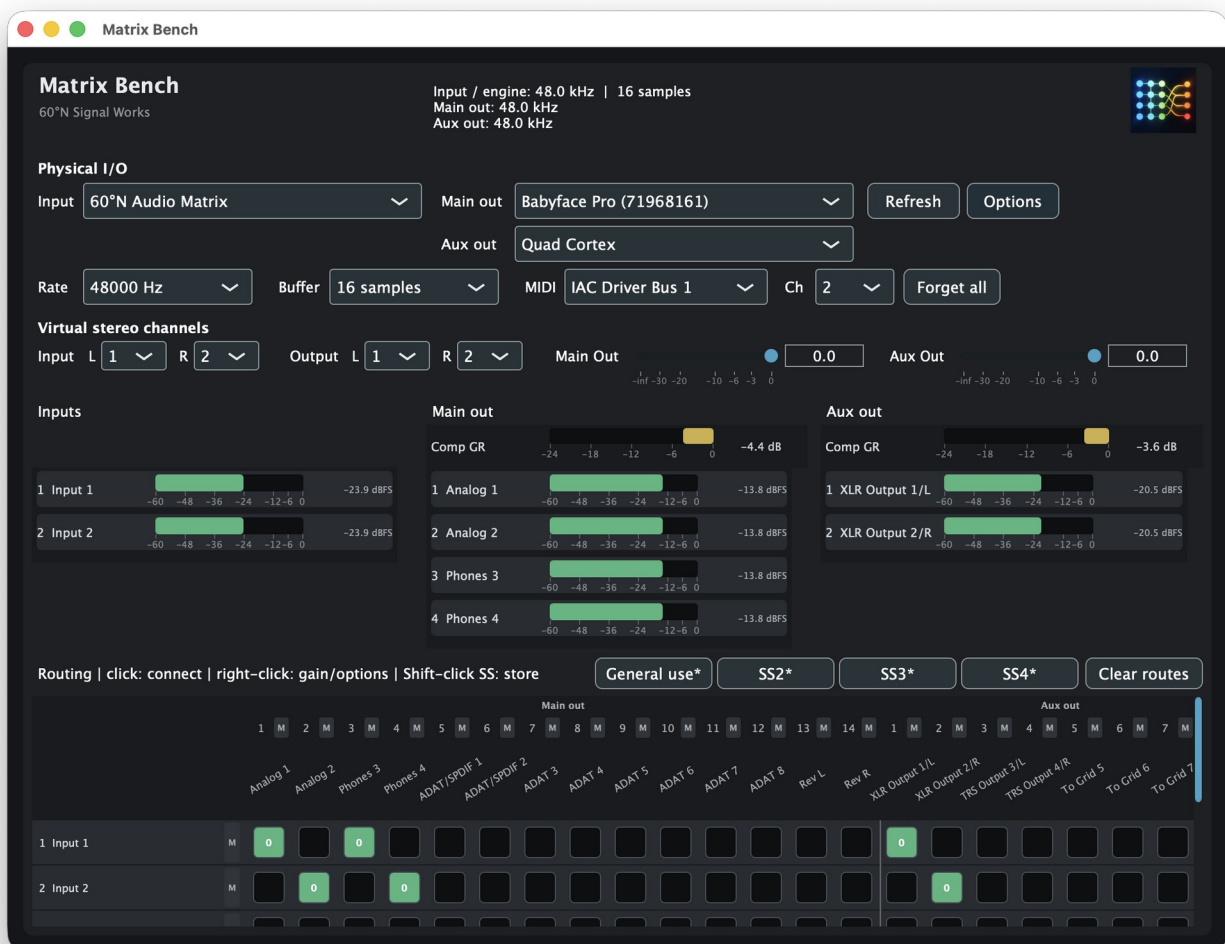
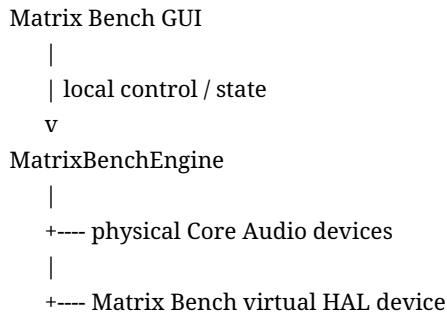


Figure 6.1. Matrix Bench 2.1.1 main routing and processing window.

6.1 Architecture

Matrix Bench is split into three cooperating parts:



Matrix Bench routing is designed to continue when the GUI window is closed. The background routing component remains active so the application window does not have to stay open for an established route.

The GUI is a controller and status surface for that engine. Closing the GUI does not intentionally stop audio.

The virtual HAL device provides eight input and eight output channels to Core Audio clients and participates in the same routing architecture.

6.2 Signal flow

For each physical or virtual input, the signal path is conceptually:

Input → raw input meter → polarity inversion → HPF / LPF → matrix crosspoints →
↳ Main: crosspoint gain → Main gain / mute → Main compressor → Main output
↳ Aux: crosspoint gain → Aux gain / mute → Aux compressor → Aux output

Input meters are intentionally taken before the per-input processing. They therefore show the incoming signal rather than the level after INV or filtering.

Each matrix crosspoint has its own gain. Main and Aux also have destination-level gain and mute controls.

6.3 Inputs

Each input channel provides:

- input metering;
- polarity inversion (INV);
- high-pass filter;
- low-pass filter;
- independent routing/gain to each matrix destination.

The input filters support 6, 12 and 24 dB/octave slopes. Processing occurs before the matrix so all destinations fed from an input receive the same selected input polarity/filter state.

This architecture avoids duplicating input conditioning independently for Main and Aux.

6.4 The crosspoint matrix

A crosspoint answers a simple question: **how much of this input is sent to this destination?**

That makes Matrix Bench suitable for both conventional routing and less conventional test/monitor arrangements. A source can feed Main, Aux, both, or neither, and the gain of each path can differ.

The matrix should be treated as the authoritative routing surface. Do not assume that selecting a device automatically creates a useful signal path; the required crosspoints must also be active.

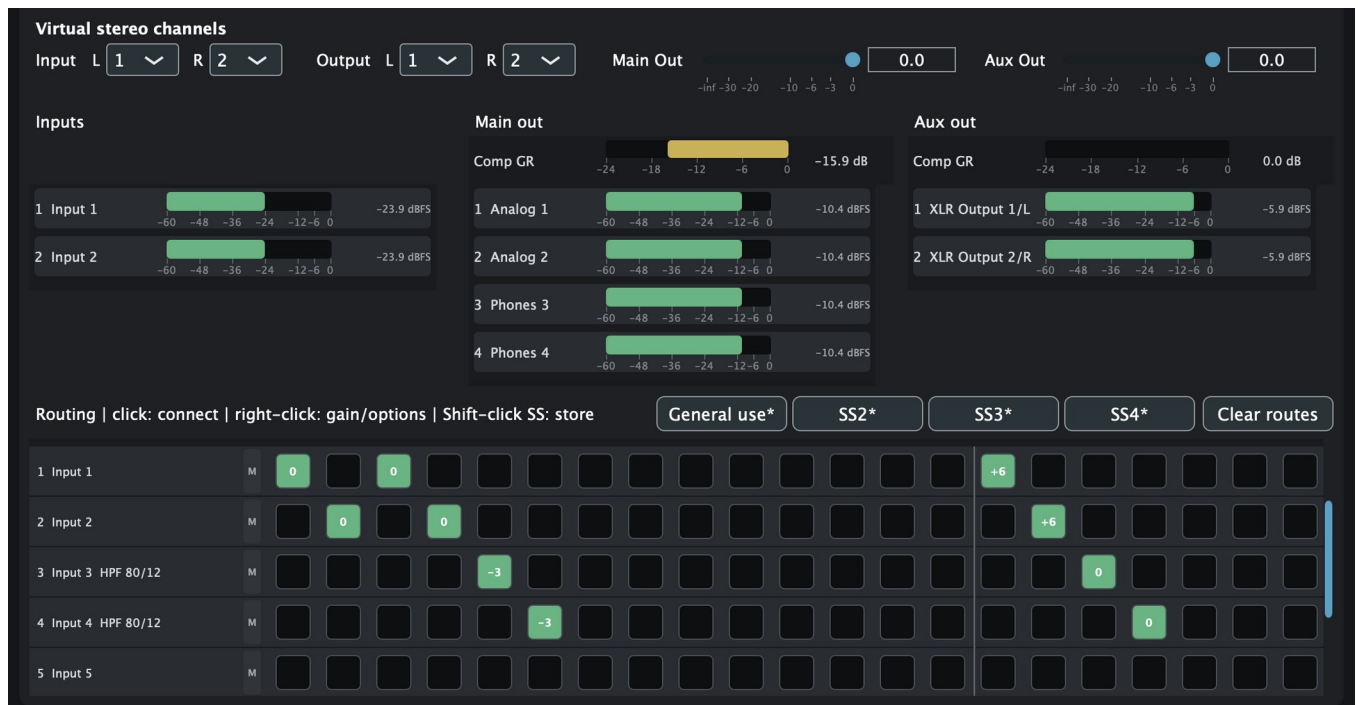


Figure 6.2. Independent input-to-destination crosspoints.

6.5 Main and Aux destinations

Main and Aux are independent first-class destinations. Each has:

- its own matrix crosspoint gains;
- destination gain;
- mute;
- output device/channel configuration;
- output metering;
- independent master compressor.

Aux is an all-channel endpoint rather than a hidden copy of Main.

This makes it possible, for example, to build one monitor mix on Main and a separate measurement, recording or alternate monitoring path on Aux without changing the input processing.

6.6 Output devices and channels

Main and Aux can target physical outputs or the Matrix virtual device, subject to feedback-prevention rules.

When changing physical devices, Matrix Bench preserves the selected channel identity where that channel exists on both devices. The purpose is to keep routing stable across device changes rather than silently reinterpreting an old channel number as a different logical destination.

If a device becomes unavailable, the engine is designed to survive the loss and recover when the device returns. Device availability is therefore a runtime condition, not a reason for the entire routing state to be discarded.

6.7 Independent output clocks and sample-rate conversion

Main and Aux may involve devices that do not share a hardware clock. Matrix Bench handles independent output clock domains rather than assuming that two selected devices remain sample-synchronous forever.

Where required, the engine performs sample-rate conversion and clock-domain adaptation between the internal routing stream and the destination device.

This matters with combinations such as a professional interface, a fixed-rate modeler and Bluetooth audio. A nominal sample-rate label alone does not establish a common physical clock.

For measurement work, prefer a single known clock domain when possible. Independent-device routing is useful, but asynchronous conversion and device buffering become part of the real system behavior.

6.8 The Matrix virtual audio device

The Matrix Bench Core Audio HAL device exposes:

- 8 inputs
- 8 outputs

to multichannel-aware macOS applications.

All eight channels are available to clients. Conventional stereo pairs 1-2, 3-4, 5-6 and 7-8 can be used by software that supports explicit multichannel selection.

The virtual device is useful for routing audio between Matrix Bench and applications without an external loopback cable.

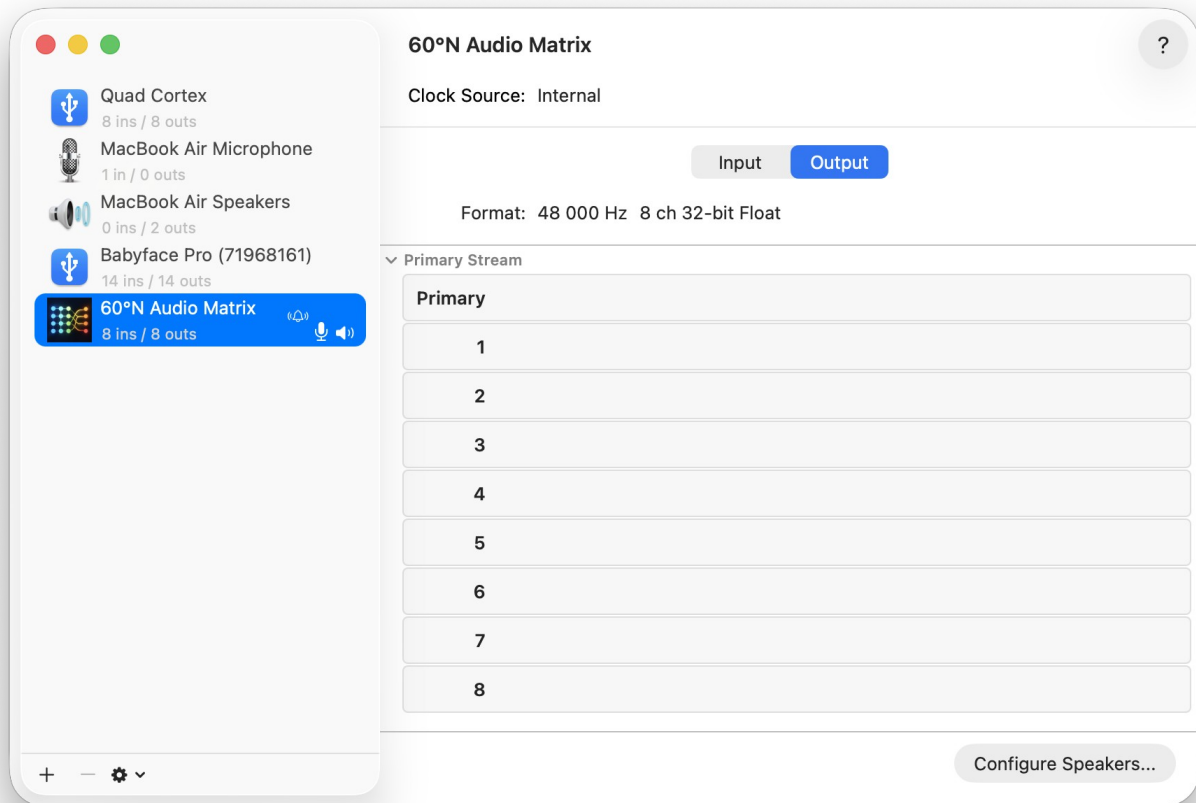


Figure 6.3. Matrix Bench 8×8 virtual Core Audio device.

6.9 macOS channel selection

The virtual device is intended to remain usable when the Matrix Bench GUI is not open. Channel configuration therefore cannot depend solely on application-local controls.

macOS-side channel selection, including non-adjacent output assignments available through Audio MIDI Setup speaker configuration, participates in the headless architecture. A configuration such as channels 4 and 8 can be selected where Core Audio exposes that mapping.

The GUI should reflect the effective OS-side configuration when it is open rather than creating a second contradictory routing truth.

For applications that are explicitly multichannel-aware, direct selection of all eight virtual channels remains preferable to relying on a stereo speaker mapping.

6.10 Feedback prevention

Virtual routing makes accidental feedback easy. Matrix Bench therefore prevents configurations in which the Matrix virtual device would be used simultaneously as both Main and Aux in a way that creates an unsafe/undefined routing loop.

A rejected device selection is intentional protection, not a hot-plug failure.

When designing a more complicated inter-application route, trace the complete signal path from source to destination before enabling crosspoints. A virtual cable is still a cable from a feedback point of view.

6.11 Headless engine

Matrix Bench includes a background routing component that starts automatically after installation. In normal use, no separate service management is required.

`works.60n.matrixbench.engine`

The GUI controls the background routing state. If the GUI cannot reach that state, reopen Matrix Bench and verify that the installed application components are available before rebuilding routes.

The important user-visible consequence is simple: **routing can continue without the GUI**.

This is particularly important for the virtual device. Core Audio clients and physical routing should not require a visible Matrix Bench application window merely to keep passing audio.

6.12 Persistence

Matrix Bench persists the routing and processing state needed to restore the working system, including:

- selected input/output devices and channels;
- matrix crosspoint gains;
- destination gains and mutes;
- input INV state;
- HPF/LPF state and settings;
- compressor state/settings;
- snapshot definitions and names;
- MIDI assignments where applicable.

Restoration is designed to preserve logical routing identity across device availability changes rather than collapsing the configuration when a device is temporarily absent.

6.13 Snapshots

Four snapshots, **SS1–SS4**, provide fast recall of routing and processing state.

Snapshot names are editable. Right-click renames a snapshot, and the UI indicates when a snapshot is being applied.

Snapshot recall applies the stored routing and processing state as one coordinated change, minimizing audible intermediate states during recall.

A snapshot can include routing, gains, mutes, INV, filters and other persisted processing state rather than being merely a matrix preset.



Figure 6.4. Matrix Bench snapshot controls.

6.14 MIDI control

Matrix Bench supports MIDI control and MIDI Learn for relevant routing/mixer actions.

The MIDI menu groups learn/clear operations and controllable parameter types such as Gain, Polarity and Mute. A global **Forget all** action clears learned assignments, and MIDI channel selection limits the control source where required.

Snapshot recall can also be learned. Option-click is used for snapshot MIDI Learn.

MIDI control changes the same engine state as GUI control; it is not a separate parallel mixer.

For diagnosing controller traffic before assigning it, MIDI Bench is the companion tool in the suite.

6.15 Main and Aux compressors

Main and Aux each have an independent studio-style master compressor. Compression is destination-specific and occurs after the matrix/destination mix.

Each compressor provides:

- bypass;
- threshold;
- attack;
- release;
- makeup gain;
- gain-reduction metering.

Detailed controls are kept behind the compressor/options interface so the main matrix remains readable.

The compressors are intended as practical output dynamics controls for music, speech and monitoring, not as hidden safety limiters. Bypass them when a measurement requires an unmodified amplitude path.

6.16 Gain-reduction metering

Each destination compressor has a gain-reduction meter distinct from the normal signal-level meter.

Gain reduction uses a negative-dB convention and is displayed top-down, making increasing compression visually different from increasing signal level. The meter is UI-smoothed independently of the audio processing.

As with the other meters, the display is a monitoring aid. The DSP state and signal path, not the meter animation, define the actual processing.

6.17 Metering

Matrix Bench provides input and destination meters with subtle engineering scales.

Input meters are pre-input-DSP. Main/Aux meters represent their destination signal path. Gain-reduction meters show compressor action.

Meter ballistics are intentionally smoothed enough to remain readable rather than reacting as raw sample-peak oscilloscopes. Meter smoothing does not alter the audio.

6.18 Buffer size

The supported buffer-size choices are:

- 16
- 32
- 64
- 128
- 256
- 512 samples

The requested buffer size and the actual callback behavior must not be confused with an arbitrary fixed HAL property.

The virtual device can use callback sizes that differ from a nominal device-property value. For the user, the important rule is not to infer complete path latency from one reported buffer property; measure the actual route when timing matters.

A buffer-size control is therefore meaningful, but it still does not by itself equal end-to-end latency.

6.19 Buffer size versus physical latency

Physical latency includes more than one callback block:

- application / engine buffering
- + driver / safety buffering
- + D/A conversion
- + external physical path
- + A/D conversion
- + any SRC / clock-domain adaptation

A 16-sample request at 48 kHz is only about 0.333 ms of sample time. It does not imply a 0.333 ms physical round trip.

Matrix Bench's virtual routing can also introduce different behavior from a direct physical path because Core Audio clients, the HAL driver and the engine participate in the route.

Use Latency Bench or the documented physical loopback procedure when the engineering question is actual path latency.

6.20 Buffer and latency confidence

Selected buffer size is only one contributor to latency. For latency-sensitive routing, measure the complete physical or virtual path rather than inferring end-to-end timing from the buffer setting.

Matrix Bench has been checked across 16, 32, 64, 128, 256 and 512 sample settings with physical audio paths. The purpose of these checks is to confirm practical routing behavior across materially different buffer conditions, not to claim that buffer size equals measured latency.

The virtual Matrix device and physical outputs can add different buffering and clock-domain behavior. When timing matters, document the actual route and measure it with Latency Bench.

6.21 Hot-plug and unavailable devices

The engine is designed for devices to disappear and return without requiring the routing configuration to be rebuilt from scratch.

When an input or Main/Aux output disappears, its logical identity is retained so that it can recover when the same device becomes available again.

For a headless system this behavior is essential. A temporary USB or Bluetooth disconnect must not require opening the GUI and manually reconstructing the entire matrix.

6.22 Device-specific considerations

Representative devices used with Matrix Bench include:

- RME Babyface Pro;
- Neural DSP Quad Cortex;
- JBL Tune 530BT;
- Mac internal audio;
- Matrix Bench virtual 8×8 device.

These devices illustrate materially different real-world constraints: the Quad Cortex operates at 48 kHz, the tested JBL Bluetooth path at 44.1 kHz with substantial wireless buffering, and the Babyface supports flexible professional-interface operation.

These are validation examples, not an exclusive compatibility list.

6.23 CLI control

The development/install architecture includes `matrixbenchctl` for command-line interaction with the engine.

It is primarily an engineering/service interface rather than the normal end-user control surface. Most users should configure routing through Matrix Bench itself or through learned MIDI control.

The handbook therefore documents the existence and architectural role of `matrixbenchctl` without making direct CLI operation a prerequisite for using the installed suite.

6.24 Installation

The intended macOS suite location is:

`/Applications/60°N Signal Works Audio Bench Suite`

Matrix Bench installation also supplies the components required by the headless engine and virtual Core Audio device.

Because the engine and HAL component outlive the GUI process, installation and removal must be treated as system integration rather than simply copying or deleting one .app bundle.

The current individual Matrix package is version 2.1.1. Suite-level installer consolidation is a separate distribution task and does not change the architecture described here.

6.25 Practical routing workflows

Independent monitor and measurement outputs

Route the required sources to Main for monitoring and create a different set of Aux crosspoints for the measurement or recording path. Keep the input DSP common unless the experiment specifically requires a different source treatment.

Feed another macOS application

Route the desired matrix destination to the Matrix virtual device and select the corresponding virtual channels in the receiving application. Verify the direction carefully and avoid routing the receiving application's return back into the same virtual path unless feedback is explicitly intended and controlled.

Recall a test configuration

Build the complete device, channel, matrix, mute, gain and processing state, save it to SS1–SS4 and give the snapshot a descriptive name. Recall should then restore the coordinated state rather than requiring a sequence of manual changes.

MIDI-controlled comparison

Use MIDI Learn for the required gains, mutes, polarity or snapshot recalls. Validate the controller messages in MIDI Bench first when the source device or channel is uncertain.

Headless always-on route

Configure and verify the route in the GUI, close Matrix Bench and confirm that audio continues through MatrixBenchEngine. Reopen the GUI to verify that it reflects the current engine state rather than starting a second independent routing session.

6.26 Troubleshooting

No audio although the device is selected

Check the matrix crosspoints as well as the device/channel selectors. A selected destination does not imply that any input is routed to it.

Device selection is rejected

Check whether the Matrix virtual device would be used in an unsafe simultaneous Main/Aux configuration. A rejection can be deliberate feedback prevention.

A device disappeared

Leave the logical configuration intact if the disconnect is temporary. The engine is designed to recover the device when it returns.

Bluetooth path feels late

Bluetooth adds its own buffering and may also force a different sample-rate/clock-domain path. A small Matrix buffer does not remove downstream Bluetooth latency.

Virtual route has more latency than expected

Verify actual client callback/buffer behavior and distinguish virtual-driver latency from physical-device round-trip latency. Use the appropriate HAL tests or Latency Bench rather than estimating from the selected buffer size alone.

Routing changes after swapping devices

Confirm that the intended channel exists on both devices and inspect the effective channel identity. Matrix Bench preserves valid channel identity rather than intentionally remapping every device to a generic first stereo pair.

6.27 Compatibility confidence

Matrix Bench has been exercised with physical interfaces, built-in audio, Bluetooth audio and its virtual 8×8 device, including device changes, persistence, snapshots, MIDI control and independent Main/Aux routing.

This is representative coverage rather than a universal compatibility guarantee. For a critical setup, verify the actual devices, channels, clocking and routing used in that measurement session.

6.28 Measurement and routing record

For a Matrix Bench configuration that needs to be reproduced, record:

- Matrix Bench version
- engine / virtual-device version where relevant
- input device and channels
- Main device and channels
- Aux device and channels
- sample rate
- buffer size
- matrix crosspoint gains
- input INV / HPF / LPF settings
- Main/Aux gain and mute
- compressor settings/bypass
- snapshot name if used
- MIDI assignments if relevant
- virtual-channel mapping
- physical wiring

For latency-sensitive work, add the measured physical or virtual path latency rather than assuming it from the selected buffer size.

7. MIDI Bench

MIDI Bench 2.0.0 for macOS is the suite's MIDI monitor, message sender and deterministic command-file runner. It is intended for practical MIDI troubleshooting and repeatable control sequences rather than music sequencing.

The application can monitor incoming MIDI with timestamps and optional raw bytes, send Channel Control Change and Program Change messages manually, and execute simple text command files with delays and looping. Device and channel selections persist between sessions.

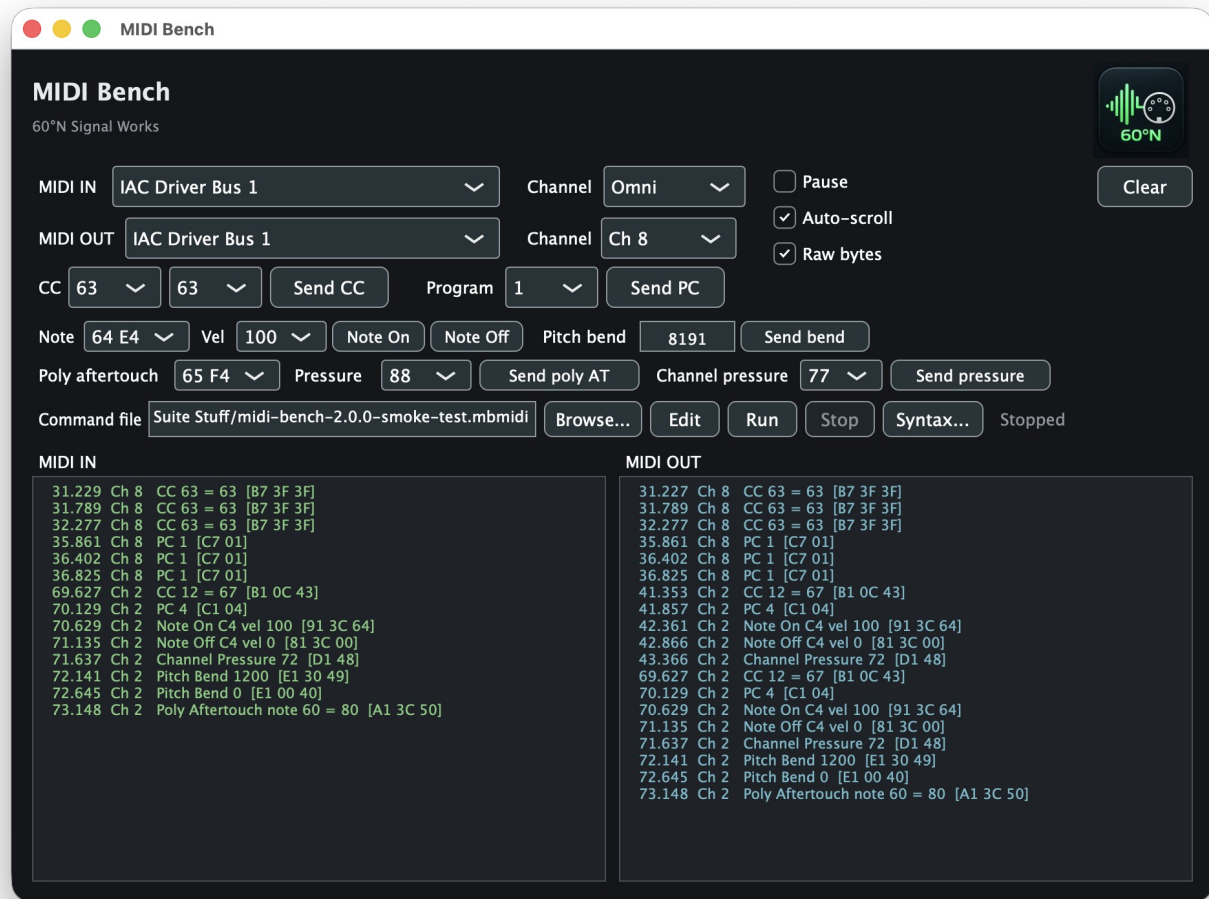


Figure 7.1. MIDI Bench 2.0.0 main window.

7.1 Typical uses

MIDI Bench is useful when the question is not "can the controller operate the target?" but **what MIDI message is actually being sent or received?**

Typical jobs include:

- identifying the channel and controller number used by a hardware control;
- checking Program Change values;
- verifying that a device is transmitting at all;
- comparing incoming and outgoing messages with timestamps;
- sending a known CC or PC message without opening a DAW;
- replaying a deterministic series of control messages;
- validating MIDI assignments before using Matrix Bench MIDI Learn.

It deliberately stays small enough to function as a bench instrument.

7.2 MIDI input monitoring

Select the required MIDI IN device and channel, then operate the source controller.

Incoming messages are shown in the **Incoming** monitor with timestamps. This makes it possible to distinguish message content from timing and order.

The monitor is particularly useful before MIDI Learn. Rather than guessing which CC a pedal, knob or switch transmits, observe it directly and then assign the known message in the target application.

7.3 MIDI output monitoring

Messages sent by MIDI Bench are shown in the **Outgoing** monitor with comparable timestamps.

Keeping Incoming and Outgoing in separate side-by-side views avoids mixing observed external traffic with messages generated by the test tool itself.

This is useful when manually reproducing an incoming controller message or when checking the sequence produced by a command file.

7.4 Monitor controls

The monitoring interface includes:

Pause

Stops live monitor updating without changing the underlying device selection.

Auto-scroll

Keeps the newest monitor traffic visible as messages arrive.

Raw bytes

Shows the underlying MIDI bytes in addition to the interpreted message information.

Clear

Clears the monitor contents so a new test can begin from a clean display.

Raw-byte display is especially useful when diagnosing a message whose interpreted representation is not enough to expose the problem.

MIDI IN	MIDI OUT
341.137 Ch 2 CC 12 = 67 [B1 0C 43]	341.136 Ch 2 CC 12 = 67 [B1 0C 43]
341.635 Ch 2 PC 4 [C1 04]	341.635 Ch 2 PC 4 [C1 04]
342.140 Ch 2 Note On C4 vel 100 [91 3C 64]	342.139 Ch 2 Note On C4 vel 100 [91 3C 64]
342.642 Ch 2 Note Off C4 vel 0 [81 3C 00]	342.642 Ch 2 Note Off C4 vel 0 [81 3C 00]
343.146 Ch 2 Channel Pressure 72 [D1 48]	343.146 Ch 2 Channel Pressure 72 [D1 48]
343.648 Ch 2 Pitch Bend 1200 [E1 30 49]	343.648 Ch 2 Pitch Bend 1200 [E1 30 49]
344.150 Ch 2 Pitch Bend 0 [E1 00 40]	344.150 Ch 2 Pitch Bend 0 [E1 00 40]
344.650 Ch 2 Poly Aftertouch note 60 = 80 [A1 3C 50]	344.650 Ch 2 Poly Aftertouch note 60 = 80 [A1 3C 50]
345.651 Ch 2 CC 12 = 67 [B1 0C 43]	345.651 Ch 2 CC 12 = 67 [B1 0C 43]
346.152 Ch 2 PC 4 [C1 04]	346.152 Ch 2 PC 4 [C1 04]
346.655 Ch 2 Note On C4 vel 100 [91 3C 64]	346.655 Ch 2 Note On C4 vel 100 [91 3C 64]
347.159 Ch 2 Note Off C4 vel 0 [81 3C 00]	347.159 Ch 2 Note Off C4 vel 0 [81 3C 00]
347.662 Ch 2 Channel Pressure 72 [D1 48]	347.662 Ch 2 Channel Pressure 72 [D1 48]
348.163 Ch 2 Pitch Bend 1200 [E1 30 49]	348.163 Ch 2 Pitch Bend 1200 [E1 30 49]
348.667 Ch 2 Pitch Bend 0 [E1 00 40]	348.667 Ch 2 Pitch Bend 0 [E1 00 40]
349.172 Ch 2 Poly Aftertouch note 60 = 80 [A1 3C 50]	349.172 Ch 2 Poly Aftertouch note 60 = 80 [A1 3C 50]
350.176 Ch 2 CC 12 = 67 [B1 0C 43]	350.176 Ch 2 CC 12 = 67 [B1 0C 43]
350.680 Ch 2 PC 4 [C1 04]	350.680 Ch 2 PC 4 [C1 04]
351.184 Ch 2 Note On C4 vel 100 [91 3C 64]	351.183 Ch 2 Note On C4 vel 100 [91 3C 64]
351.688 Ch 2 Note Off C4 vel 0 [81 3C 00]	351.688 Ch 2 Note Off C4 vel 0 [81 3C 00]
352.192 Ch 2 Channel Pressure 72 [D1 48]	352.192 Ch 2 Channel Pressure 72 [D1 48]
352.696 Ch 2 Pitch Bend 1200 [E1 30 49]	352.696 Ch 2 Pitch Bend 1200 [E1 30 49]
353.200 Ch 2 Pitch Bend 0 [E1 00 40]	353.200 Ch 2 Pitch Bend 0 [E1 00 40]
353.730 Ch 2 Poly Aftertouch note 60 = 80 [A1 3C 50]	353.730 Ch 2 Poly Aftertouch note 60 = 80 [A1 3C 50]

Figure 7.2. Timestamped Incoming and Outgoing MIDI monitoring.

7.5 MIDI channels

MIDI channel selection is explicit. A device can be physically present and active while still appearing ineffective if the source and target are listening on different channels.

When troubleshooting:

1. identify the source device;
2. verify the channel;
3. verify the message type;
4. verify its number/value;
5. only then configure MIDI Learn or the destination device.

This order avoids treating a channel mismatch as a routing or application failure.

7.6 Manual sending

MIDI Bench can send Channel Control Change (**CC**) and Program Change (**PC**) messages manually.

Manual sending is useful for isolating the destination from the physical controller. If a known CC sent from MIDI Bench operates the target but the hardware controller does not, the problem is upstream of the target's MIDI handling.

Conversely, if the known message leaves MIDI Bench correctly but the target does not respond, inspect the target device, MIDI port, channel and assignment.

7.7 Control Change

A Control Change message identifies a MIDI channel, controller number and value.

The command-file representation is:

```
CH 1 CC 12 VAL 67
```

This sends CC number 12 with value 67 on MIDI channel 1.

For manual use, select the corresponding channel, CC number and value in the Send controls and transmit the message.

7.8 Program Change

A Program Change identifies a MIDI channel and program number.

The command-file representation is:

```
CH 3 PC 4
```

This sends Program Change 4 on MIDI channel 3.

Remember that hardware user interfaces do not all label programs with the same zero-/one-based convention. MIDI Bench shows and sends the actual command value used by its interface/file syntax; verify the receiving device's mapping rather than assuming its displayed preset number is identical.

7.9 Run from file

Version 2.0.0 adds a small deterministic command-file sequencer.

A command file is ordinary text. It can contain MIDI commands and explicit delays, and can be run once or repeatedly.

The controls are:

- **Browse**, select a .mbmidi command file;
- **Edit**, open it explicitly in macOS TextEdit;
- **Run**, validate and execute it;
- **Stop**, stop execution;
- **Syntax**, show the compact command reference.

The application remembers the last command-file location.

The .mbmidi extension is required by the validated command-file format.

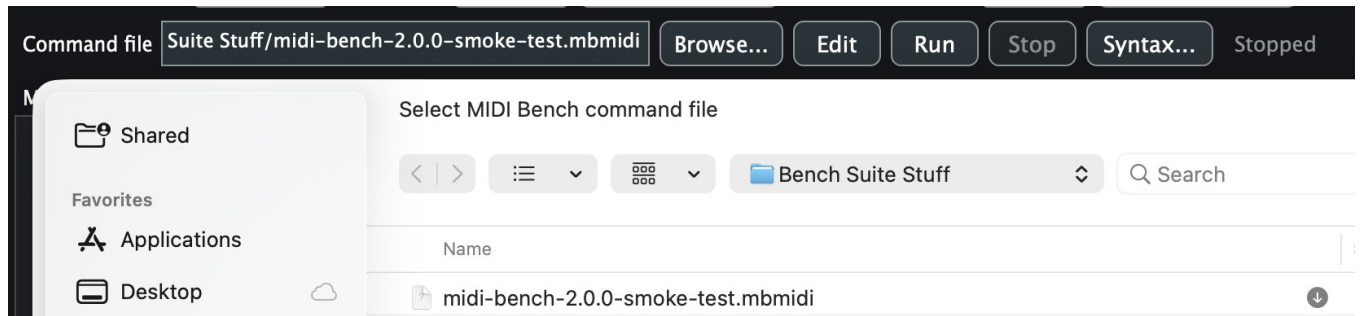


Figure 7.3. Run-from-file sequencer controls.

7.10 Command-file syntax

The command-file format is UTF-8 text with the required header:

```
MIDI-BENCH-FILE 1.0
```

Blank lines and # comments are allowed. Supported commands are:

```
CH <1..16> CC <0..127> VAL <0..127>
CH <1..16> PC <0..127>
CH <1..16> NOTE <0..127> VEL <0..127> ON
CH <1..16> NOTE <0..127> OFF
CH <1..16> BEND <-8192..8191>
CH <1..16> PRESSURE <0..127>
CH <1..16> POLYAT <0..127> VAL <0..127>
WAIT <0..3600000>
LOOP
```

WAIT is specified in milliseconds. LOOP, when used, must be the final command and restarts execution from the beginning until **Stop** is pressed.

The whole file is validated when selected and revalidated from disk immediately before every Run.

7.11 Validation before execution

MIDI Bench validates the whole command file when it is selected and reloads and revalidates it from disk immediately before every Run.

It rejects the wrong extension, a wrong or missing header, malformed commands, out-of-range values, commands after LOOP, and files larger than 1 MiB. A rejected file reports Invalid file (line N) together with a detailed line-numbered diagnostic.

After a failed validation, **Run remains available while the selected file still exists**. This supports **Edit -> fix -> save -> Run** without browsing for the file again. Run never executes a partially understood file: the corrected file must pass complete revalidation before execution begins.

7.12 Editing command files

Edit opens the selected command file in the default text editor rather than requiring a dedicated MIDI Bench editor.

This keeps the file format transparent and makes command files easy to store in Git, compare, copy or generate with other tools.

After editing, return to MIDI Bench and let it revalidate the file before running it.

7.13 Looping

Enable **Loop** when the entire valid command sequence should repeat.

Looping is useful for:

- repeatedly switching a DUT between two states;
- cycling Program Changes during observation;
- exercising a MIDI-controlled function for stability testing;
- creating a repeatable control pattern while another Bench measures the audio result.

Use Stop to terminate execution.

A looped command file is still a control sequence, not a real-time musical sequencer. Its purpose is deterministic bench operation.

7.14 Persistence

MIDI Bench persists the selected MIDI IN device, MIDI OUT device and channel configuration so routine bench setups do not need to be reconstructed on every launch.

Persistence should not be confused with device availability. A previously selected USB MIDI device can remain the intended configuration while being physically disconnected.

When troubleshooting after reconnecting hardware, first verify that the intended device is actually available before changing a previously correct MIDI mapping.

7.15 Deterministic MIDI sender

The project includes a deterministic MIDI sender used for validation and test work. Its role is to produce repeatable known traffic so monitoring and parsing behavior can be checked independently of a human-operated controller.

This complements, rather than replaces, physical-controller testing. A deterministic sender proves how MIDI Bench handles known input; the real controller test proves what the external device actually emits.

7.16 Working with Matrix Bench

MIDI Bench and Matrix Bench are natural companions.

A reliable MIDI Learn workflow is:

1. connect the physical controller;

2. open MIDI Bench and identify its MIDI device;
3. operate the intended control;
4. verify the actual channel, CC/PC number and value behavior;
5. open Matrix Bench;
6. use MIDI Learn for the intended gain, polarity, mute or snapshot action;
7. test the learned action;
8. return to MIDI Bench if the behavior differs from what was expected.

This separates **message diagnosis** from **application assignment**.

For snapshot recall, Matrix Bench supports learned MIDI control, including its Option-click learning workflow. MIDI Bench can establish exactly what the controller sends before that assignment is created.

7.17 Working with audio measurements

MIDI command files can also coordinate repeatable DUT states while Signal, Spectral or Latency Bench handles the audio side.

For example:

```
set DUT to state A
wait
perform or observe measurement A
```

```
set DUT to state B
wait
perform or observe measurement B
```

MIDI Bench itself does not automatically trigger or synchronize the other Bench applications. The useful property is that the MIDI-side state changes can be made repeatable.

When exact audio/MIDI synchronization matters, measure or otherwise establish the timing rather than assuming that a MIDI command and the resulting audio state change are simultaneous.

7.18 Troubleshooting

No incoming messages

Confirm the MIDI IN device, physical connection and channel. Then enable Raw bytes to see whether traffic is present but not represented as the message type you expected.

The target does not respond to a manual send

Confirm MIDI OUT, channel and the target's expected CC/PC mapping. Compare the outgoing message with a known working incoming controller message when possible.

Run is disabled

The selected command file is invalid or has become invalid after editing. Correct the file and allow MIDI Bench to revalidate it.

Edit opens an unexpected application

The command file is UTF-8 text with a .mbmidi extension. **Edit** opens it explicitly in macOS TextEdit, so no .mbmidi file association is required.

A loop changes the DUT too quickly

Add explicit WAIT delay commands. Hardware can require non-zero settling time after Program Change or other state changes.

Program number looks off by one

Check the receiving device's display convention. Do not compensate blindly until the actual MIDI Program Change value has been verified.

7.19 Measurement confidence

MIDI Bench 2.0.0 has been checked for the following user-visible functions:

- MIDI input device and channel selection;
- timestamped incoming monitoring;
- MIDI output selection and manual CC/PC sending;
- timestamped outgoing monitoring;
- Pause, Auto-scroll, Raw bytes and Clear;
- persistence of IN/OUT device and channel;
- deterministic test traffic;
- command-file Browse/Edit/Run/Stop/Syntax operation;
- CC, PC, NOTE, BEND, PRESSURE and POLYAT command parsing;
- delay commands;
- validation and rejection of invalid files;
- Edit -> fix -> save -> Run revalidation without re-browsing;
- Edit using macOS TextEdit;
- remembered command-file location.

The tests establish application behavior with the tested MIDI environment. They do not establish the response time, numbering convention or implementation quality of an arbitrary external MIDI device.

7.20 Recording a MIDI test

For a reproducible MIDI-side test, record:

- MIDI Bench version
- MIDI IN device
- MIDI OUT device
- channel
- message type
- CC / PC number
- value where applicable
- command-file contents and revision
- delay settings
- Loop state
- target device and target function
- observed incoming/outgoing timestamps

When MIDI is controlling an audio measurement, record the corresponding audio Bench configuration and DUT state as part of the same test record.

8. Cross-Bench measurement workflows

The five Bench applications are most useful as a coordinated toolkit. A repeatable measurement normally separates four jobs: **generate a known stimulus, route it deliberately, observe the audio result, and control or document DUT state.**

Signal Bench provides defined stimuli. Spectral Bench measures level, spectrum, distortion, frequency response and phase. Latency Bench measures physical-path timing. Matrix Bench handles routing, monitoring and repeatable audio configurations. MIDI Bench identifies and reproduces MIDI-side state changes.

No workflow requires every application. Use the smallest combination that answers the engineering question.

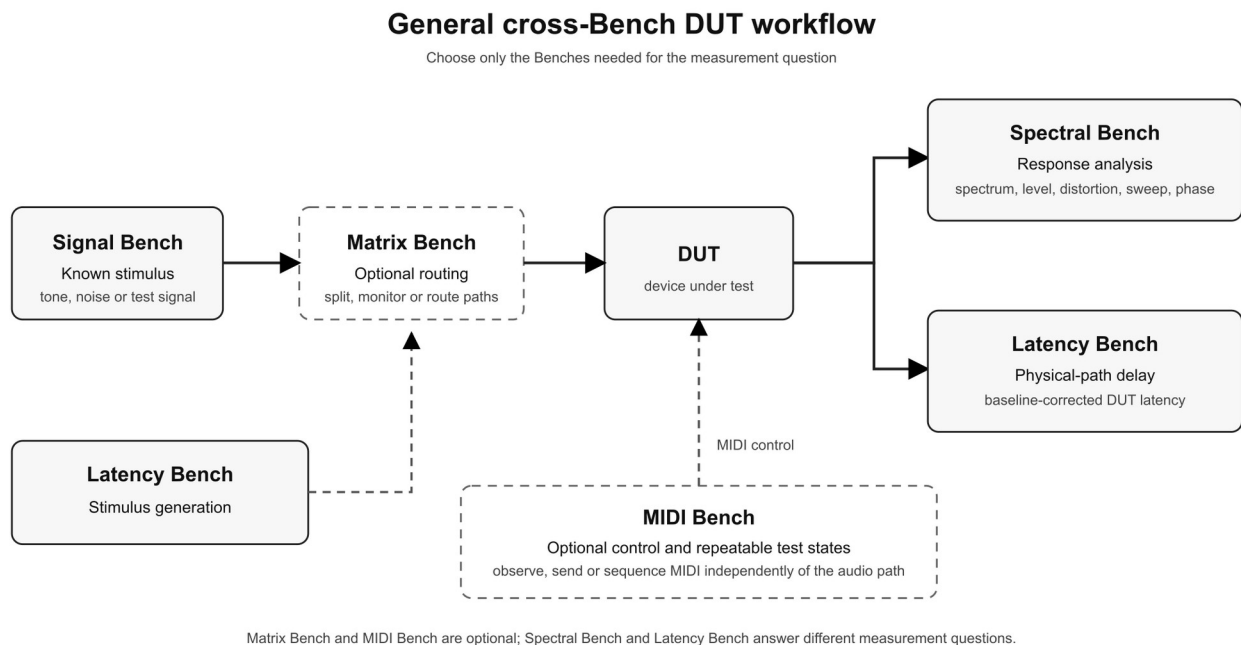


Figure 8.1. General cross-Bench DUT measurement workflow.

8.1 Start with the question

Before connecting software or cables, define the quantity being measured.

Examples:

- frequency response relative to a reference;
- harmonic distortion at a specified tone and level;
- selective intermodulation products;
- physical-path latency;
- timing of a filtered subwoofer output;
- difference between two DUT states;
- MIDI message behavior;
- latency added by an effects loop;
- routing behavior through a virtual or physical path.

This determines which Bench owns the authoritative result. A Spectral Bench phase trace is not a substitute for a Latency Bench physical-path latency measurement, and a Latency Bench complete-response timing result is not a THD or frequency-response measurement.

8.2 Establish the common test conditions

Record the conditions that can change the result before comparing measurements:

- sample rate
- audio interface and channels
- buffer size where relevant
- stimulus type and level
- DUT input/output levels
- DUT state
- physical cabling
- reference path
- Matrix routing/processing state
- MIDI program/control state

Use conservative levels first. Clipping in the generator, interface, DUT or return path invalidates many otherwise sophisticated measurements.

When comparing A and B, change only the variable that is meant to differ.

8.3 Signal Bench + Spectral Bench: level and spectrum

For a basic spectral test:

1. select a defined Signal Bench stimulus;
2. set a conservative Output Level;
3. route it through the DUT;
4. select the DUT return in Spectral Bench;
5. begin with Input Gain at 0 dB;
6. choose the spectrum viewport and averaging needed for readable inspection;
7. save the result when the setup is stable.

Use Pink or White noise for broadband inspection. Noise Smooth may improve the visual readability of broadband noise, but it does not change Spectral Bench's authoritative numeric measurement data.

For a discrete tone, use a matching defined measurement mode where possible.

8.4 Signal Bench + Spectral Bench: single-tone distortion

Signal Bench and Spectral Bench share defined 100 Hz, 1 kHz and 10 kHz measurement cases.

A repeatable distortion workflow is:

1. select the matching Signal Bench tone preset;
2. set the desired generator Output Level;
3. verify DUT and return-path headroom;
4. select the corresponding Spectral Bench measurement mode;
5. confirm that the fundamental is valid;
6. inspect H2-H10 and THD;
7. inspect THD+N and record its measurement bandwidth;
8. save PNG, CSV and TXT result companions.

Do not compare THD+N values without preserving the bandwidth definition. Do not infer analog voltage from dBFS without the interface/DUT calibration required to establish that relationship.

8.5 Signal Bench + Spectral Bench: intermodulation

For the suite's defined two-tone tests, use matching generator and analyzer modes.

CCIF / DFD-style

Signal Bench generates equal-amplitude 19 kHz + 20 kHz tones. Spectral Bench inspects the defined 1 kHz, 18 kHz and, where valid below Nyquist, 21 kHz products.

SMPTE-style

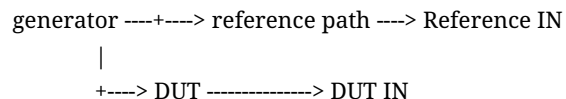
Signal Bench generates 60 Hz + 7 kHz with a 4:1 low/high amplitude ratio. Spectral Bench inspects the defined second- and third-order sideband pairs.

The suite intentionally reports these as selective **CCIF / DFD-style** and **SMPTE-style** measurements. Do not turn the reported products into a claimed normative aggregate percentage unless that separate standard-defined calculation has actually been performed.

8.6 Referenced DUT frequency response

A referenced Spectral Bench sweep is preferred when the goal is the DUT's transfer response rather than the absolute response of the complete capture chain.

Conceptually:



First run a direct dual-channel loopback. This qualifies the channel pair and reveals residual interface/channel mismatch.

Then insert the DUT only in the DUT path and run **Transfer**. The result represents the DUT path relative to the simultaneously measured reference.

Use **Actual** instead when the captured DUT response itself is the intended quantity.

Do not call an unreferenced Actual trace a transfer function with the interface response removed.

8.7 Phase comparison

Choose the phase reference according to the question.

Raw retains transport delay and all measured phase.

Auto removes a response-dependent estimated constant-delay component. It is useful for viewing residual phase but is not an independent latency measurement.

Manual removes a known delay entered by the user. This is the appropriate cross-Bench mode when Latency Bench has independently established the physical transport delay and Spectral Bench is being used to inspect phase after that delay is removed.

Baseline is useful for A/B work. Store state A as the phase baseline, change only the DUT state, then measure state B. Common path delay is removed and the phase change between the two states remains.

8.8 Spectral Bench + Latency Bench: separate phase from physical latency

This is one of the most important cross-Bench distinctions.

A transfer-function phase measurement cannot always distinguish pure transport delay from an equivalent linear phase term belonging to the DUT response. Latency Bench attacks a different problem by measuring the relative arrival of a deterministic physical-path probe.

A useful workflow is:

1. use Latency Bench to measure the physical-path latency in an appropriate broadband/reference DUT state;
2. record the result in samples and milliseconds;
3. perform the Spectral Bench referenced sweep;
4. select Manual phase;
5. enter the independently measured delay;
6. interpret the remaining phase as the response after that chosen constant-delay component has been removed.

In a physical example, a path measured by Latency Bench at 89.15 samples / 1.857 ms at 48 kHz behaved as expected when that independently measured delay was entered in Spectral Bench Manual phase.

Auto phase compensation need not equal the Latency Bench value, and a difference between them is not by itself an error.

8.9 Filter and crossover characterization

A filter changes both magnitude and phase. Its complete-response timing can therefore differ substantially from the device's broadband transport timing.

A useful characterization sequence is:

1. measure a broadband/bypass physical latency with Latency Bench where possible;
2. run a referenced Spectral Bench sweep with the filter bypassed;
3. enable the filter without changing the rest of the path;
4. run a second sweep;
5. use Baseline phase for a direct Off/On phase comparison;
6. measure the filtered physical path with Latency Bench if arrival timing matters;
7. report the filtered timing as complete-response timing unless the measurement specifically isolates transport.

For a steep low-pass or subwoofer output, a several-millisecond increase can be physically reasonable because filter group delay contributes to the measured arrival.

8.10 Strongly bandwidth-limited and subwoofer outputs

Do not widen a legitimate subwoofer/crossover output merely to satisfy an assumed latency-test bandwidth rule.

Latency Bench 1.1.1 automatically moves to extended analysis when normal broadband analysis has weak or competing timing evidence. Extended analysis uses a longer probe and magnitude-matches the reference spectrum to the captured DUT magnitude while leaving DUT phase intact.

Strongly filtered 20-160 Hz paths can still be measured when sufficient timing evidence remains. If the evidence is ambiguous, rejection is the correct result.

For the same DUT, Spectral Bench can characterize the transfer magnitude and phase while Latency Bench answers the physical arrival-time question.

8.11 Effects-loop latency

An analog effects loop inside an otherwise digital device can add measurable latency even when no external effect is connected.

A clean comparison is:

1. establish the Latency Bench baseline;
2. measure the DUT in a state that bypasses the loop path;
3. enable or route through one loop and measure again;
4. repeat for additional loops or routing states;
5. compare medians while keeping the interface, sample rate, cabling and DUT processing otherwise unchanged.

If the loop path contains D/A and A/D conversion, most of its latency can remain present whether an external effect is active or bypassed. Distinguish **routing through the loop path** from **enabling processing inside an external effect**.

Use Spectral Bench as a companion if the loop also changes frequency response, level, phase or distortion.

8.12 DSP-block latency

To investigate whether a DSP block changes latency:

1. choose a DUT state with the block absent or bypassed;
2. measure physical-path latency;
3. enable the block without changing unrelated routing;
4. repeat the measurement;
5. compare the run statistics, not just one displayed number.

Some devices retain the same processing topology when a block is bypassed, so the block's apparent latency may not disappear when its audible processing is switched off. The measurement should describe the tested states rather than assuming what "bypass" means internally.

For filters and other phase-shifting blocks, separate transport changes from complete-response timing changes.

8.13 Matrix Bench as the routing layer

Matrix Bench becomes useful when a test requires more routing than one generator-to-DUT-to-analyzer path.

Examples include:

- feeding Main monitors while Aux carries a measurement path;
- sending one source to multiple destinations;
- feeding another macOS application through the virtual 8×8 device;
- recalling complete test routing with a snapshot;
- maintaining a headless route while measurement applications are opened and closed.

Keep measurement-path processing explicit. Disable Matrix input filters or Main/Aux compression when they are not part of the DUT being measured.

Record crosspoint gains, destination gains, mutes and any active processing with the measurement.

8.14 Matrix Bench virtual routing

The Matrix virtual device can replace physical loopback wiring for inter-application routing, but it changes the path being measured.

A virtual path can include:

Source application → virtual audio path → Matrix Bench routing → destination application or physical output

Do not assume that its latency equals a direct physical interface path or that selecting a 16-sample buffer makes the complete path 16 samples long.

When latency matters, measure the actual topology. When only signal routing matters, verify channels and guard against virtual feedback.

8.15 Matrix snapshots for repeatable A/B routing

Snapshots are useful when A/B comparison requires coordinated routing changes.

Build state A completely, save it to one snapshot, then build and save state B. Give both snapshots descriptive names. Recall them while observing the DUT or analyzer.

Snapshot recall is coordinated so intermediate routing states are minimized. Even so, allow the DUT and measurement analyzer to settle before treating the next measurement as valid.

If the A/B question can be answered by changing only one DUT parameter, prefer changing only that parameter. Snapshots are most valuable when the complete routing state genuinely needs to change.

8.16 MIDI Bench for deterministic DUT states

Use MIDI Bench when DUT state is controlled by MIDI.

First monitor the physical controller and establish the actual channel, CC/PC number and value behavior. Then either use the verified controller directly or encode the required state changes in a command file.

A command file is particularly useful when the same A/B sequence will be repeated many times. Include explicit WAIT delays when the hardware needs time to switch or settle.

MIDI Bench does not synchronize the audio measurement applications automatically. Treat MIDI command time and audible DUT-state-change time as separate unless their relationship has been established.

8.17 MIDI-controlled Spectral Bench comparison

For a MIDI-controlled filter, preset or processing state:

1. verify the MIDI command in MIDI Bench;
2. set state A;
3. wait for the DUT to settle;
4. save the Spectral Bench result or phase baseline;
5. send the state-B command;
6. wait the same defined settling interval;
7. measure state B;
8. save the MIDI command file with the audio results.

This gives the measurement an explicit control-state record rather than relying on a remembered footswitch sequence.

8.18 MIDI-controlled latency comparison

The same principle applies to Latency Bench, but each DUT measurement already consists of repeated automatic runs.

Set the DUT state first, allow it to settle, then start the ten-run latency measurement. Do not switch DUT state during those runs.

For state B, perform the complete state change before starting the next measurement. Compare the resulting medians and spread.

If a state change also alters bandwidth or phase strongly, expect complete-response timing and estimator evidence to change as well.

8.19 Sample-rate and clock-domain discipline

Keep the complete path in one sample-rate/clock domain when the experiment permits it.

A fixed-48 kHz DUT, a 44.1 kHz Bluetooth destination and a variable-rate professional interface are not interchangeable measurement paths. Matrix Bench can bridge independent output clocks, but sample-rate conversion and asynchronous buffering then become part of the real topology.

Record the actual sample rate used by the measurement application and the relevant DUT/interface constraints.

When comparing latency in samples, remember that the duration of one sample changes with sample rate. For cross-rate comparisons, milliseconds are usually the clearer physical quantity.

8.20 Buffer-size experiments

When studying buffer size, change the buffer and nothing else.

A useful sequence is:

16 / 32 / 64 / 128 / 256 / 512 samples

At each size, verify that the device/client actually accepted and is using the requested buffer. Then measure the real path.

Do not derive end-to-end latency by multiplying one selected buffer by sample time. Driver safety buffers, converters, engine buffering, DSP and other components can contribute independently.

For buffer-size experiments, keep the complete route and measurement method constant while changing only the requested buffer size. Then measure the actual physical or virtual path rather than assuming the requested value is the complete latency.

8.21 Bluetooth and wireless monitoring

Bluetooth is useful for listening but is usually a poor reference path for low-latency measurement.

Its buffering can dominate the total delay, and the tested JBL path also operates at a fixed 44.1 kHz. If a measurement is intended to characterize a DUT rather than Bluetooth itself, keep the wireless path out of the measurement loop.

If Bluetooth is the system under test, measure the complete Bluetooth path and describe it as such.

8.22 Repeating a difficult measurement

After an unusual or difficult DUT measurement, return to a known reference.

For example:

1. measure direct/bypass;
2. measure the difficult filtered or routed state;
3. restore direct/bypass;
4. measure it again.

If the final reference no longer matches the first, investigate routing, sample rate, device state, cabling or baseline validity before trusting the middle result.

This simple "reference, DUT, reference" pattern is one of the strongest defenses against an unnoticed setup change.

8.23 Saving a complete measurement set

Keep the machine-readable result together with the human-readable setup record.

For a spectral test, retain the PNG, CSV and TXT companions. For latency work, record the run statistics and evidence together with the baseline and path definition. For MIDI-controlled tests, retain the command file. For Matrix-routed tests, record or snapshot the routing state.

A useful directory layout is:

```
measurement-name/  
  README.txt  
  spectral-result.png  
  spectral-result.csv  
  spectral-result.txt  
  midi-state.mbmidi  
  routing-notes.txt  
  latency-notes.txt
```

Not every test produces every file. The point is to keep the evidence and the setup definition together.

8.24 Minimum measurement record

A measurement that may need to be repeated later should make the following recoverable:

```
question being measured  
Bench application(s) and versions  
audio device/interface  
sample rate  
buffer size if relevant  
physical and/or virtual routing  
stimulus and level  
DUT identity and state  
reference/baseline method  
measurement mode and bandwidth  
result and repeatability/statistics  
MIDI state/sequence where relevant  
anything intentionally bypassed or compensated
```

This is more valuable than adding decimal places to a result whose path definition is unknown.

8.25 Choosing the authoritative Bench

When several Bench applications are involved, keep one owner for each quantity:

Quantity	Authoritative tool
Generated stimulus definition	Signal Bench
Spectrum / level / harmonics / THD / THD+N / IM products	Spectral Bench
Referenced magnitude and phase response	Spectral Bench
Physical-path latency / complete-response timing	Latency Bench
Audio routing and processing state	Matrix Bench
MIDI message and command sequence	MIDI Bench

Cross-checking between tools is valuable, but do not silently redefine a quantity because another tool displays something that looks numerically similar.

9. Measurement confidence and validation

A measurement application can be well tested and still be used in an invalid setup. This chapter separates confidence in the Bench itself from confidence in the measurement you are making.

9.1 What validation means to the user

The Audio Bench Suite is checked with several complementary kinds of evidence: known-answer signals, physical loopbacks, representative real hardware, difficult or ambiguous cases, and repeatability checks. No single test method covers the whole measurement chain.

This supports confidence that the documented functions behave as intended under tested conditions. It does not guarantee that every interface, driver, cable, DUT, sample rate or routing arrangement will produce a valid result.

9.2 Instrument confidence versus measurement confidence

Instrument confidence concerns the Bench: whether its generator, analyzer, routing or estimator behaves according to its documented definition. Measurement confidence concerns one particular result: whether the physical path, channels, levels, sample rate, reference and DUT state were correct for that measurement.

A qualified application cannot make a clipped signal, wrong channel, stale baseline or unintended route valid. Always verify the measurement setup independently.

9.3 Use a known reference

A known reference is one of the strongest ways to distinguish DUT behavior from setup error. Examples include direct cable loopback, dual-channel loopback before a referenced sweep, a Latency Bench baseline, a Spectral Bench phase baseline, or a Matrix route with processing bypassed.

A reference remains valid only while the conditions that define it remain unchanged. Re-establish it after relevant changes to interface, sample rate, channels, wiring, gain, routing or clocking.

9.4 Reference → DUT → reference

For a difficult or surprising measurement, measure the known reference, insert the DUT or unusual condition, then return to the known reference. If the final reference no longer reproduces the initial one, investigate the setup before accepting the DUT result.

9.5 Repeatability is not accuracy

Repeated results are useful evidence of stability, but a systematically wrong setup can repeat perfectly. Repeatability becomes much stronger evidence when combined with a known reference, an independently calculated value, a second method, or a predictable change in the DUT.

Where a Bench reports statistics, use them to describe run-to-run stability. Do not treat low variation by itself as proof of absolute accuracy.

9.6 Rejection can be the correct result

A trustworthy measurement tool should sometimes refuse to produce a definitive result. Latency Bench can reject ambiguous competing timing candidates; MIDI Bench rejects invalid command files; Matrix Bench can reject unsafe virtual routing; clipped or otherwise invalid spectral measurements should not be treated as authoritative merely because numbers remain visible.

Do not work around a rejection simply to obtain the expected number. Correct the setup if possible; otherwise report the limitation.

9.7 What has been checked

Signal Bench has been checked for signal-generation, level, filtering and noise behavior across relevant sample rates. Spectral Bench has been checked with known spectral cases, physical loopback and representative DUT measurements. Latency Bench has been checked with known delays, physical paths, filtered paths, polarity inversion and ambiguous cases. Matrix Bench has been exercised with physical and virtual routing, device changes and representative buffer conditions. MIDI Bench has been checked for monitoring, sending and deterministic command-file operation.

These summaries describe the basis for confidence in the released tools without turning a user measurement into a software test procedure.

9.8 What validation does not guarantee

Validation does not guarantee compatibility with every audio or MIDI device, identical behavior on every operating-system release, identical latency from every driver, or correctness after an unrecorded change in routing, sample rate or gain. It also cannot validate an arbitrary third-party DUT.

The scope of a measurement claim should not exceed the evidence supporting it.

9.9 A practical confidence checklist

1. Establish the simplest known reference path.
2. Verify devices, channels, sample rate, levels and clocking.
3. Confirm that the measurement is not clipped and that the intended DUT is actually in the path.
4. Record or store the baseline/reference when the Bench supports one.
5. Repeat the measurement and inspect its spread or stability.
6. For a surprising result, return to the reference before changing the DUT.
7. Report rejected or inherently variable measurements as such rather than forcing a stronger conclusion.

10. Troubleshooting and measurement pitfalls

Troubleshooting a measurement system is easiest when the problem is divided into layers. Avoid changing several settings at once. Start with the simplest known path, establish where the expected signal or behavior stops, then add complexity one element at a time.

A useful order is:

Physical connection → device and channel selection → sample rate / clocking
→ level and mute state → application routing → DUT state → measurement mode → interpretation

When a result is surprising but plausible, verify the setup before explaining the number.

10.1 No signal

Work from the physical path inward.

Check:

1. source/generator is actually running;
2. correct physical output is connected;
3. DUT input and output are active;
4. correct interface input is selected;
5. interface mixer is not muting or rerouting the return;
6. correct application input/channel is selected;
7. Matrix crosspoint exists if Matrix Bench is in the path;
8. Main/Aux mute and gain are correct;
9. the DUT has not changed state through MIDI or preset recall.

A device being visible in a selector does not prove that the intended channel carries signal.

10.2 Implausibly low level

A low reading can be real, but first exclude accidental attenuation.

Inspect:

- Signal Bench Output Level;
- interface output and input gain;
- DUT input/output controls;

- Matrix crosspoint gain;
- Matrix Main/Aux destination gain;
- active filters;
- Spectral Bench Input Gain;
- balanced/unbalanced wiring where relevant.

Spectral Bench Input Gain changes the samples entering the analyzer. Its dB viewport controls do not. Do not use viewport movement to compensate for a physically weak measurement signal.

10.3 Clipping

Clipping invalidates many spectral and distortion measurements and can also degrade correlation-based timing.

Look for clipping at every stage, not only the final application meter:

Generator → interface output → DUT input → DUT processing → DUT output
 → interface input → Matrix processing (if used) → analyzer

Reduce stimulus or upstream gain until the complete path has adequate headroom.

A lower clean signal is generally more useful than a louder clipped one.

10.4 Noise floor is unexpectedly high

First determine whether the noise belongs to the DUT, the analog path or the measurement configuration.

Useful checks are:

- terminate or disconnect the DUT input as appropriate;
- compare direct loopback with the DUT path;
- reduce unnecessary analog gain;
- check for ground or cabling problems;
- bypass processing not under test;
- verify that Spectral Bench Input Gain is not being mistaken for a physical SNR improvement;
- compare left/right or reference/DUT channels where the setup permits.

For THD+N, preserve the measurement bandwidth. A noise number without its bandwidth definition is incomplete.

10.5 Wrong frequency or missing high-frequency content

Check the sample rate first.

Nyquist frequency is half the sample rate. A requested or expected component above Nyquist cannot appear as a valid in-band component.

Also inspect:

- DUT filters;
- Signal Bench broadband HPF/LPF;
- Matrix input HPF/LPF;
- fixed-rate hardware;
- sample-rate conversion in the path;
- the selected Spectral Bench frequency range.

For harmonic measurements, higher harmonics can legitimately fall outside the measurable band.

10.6 Sample-rate mismatch

Different devices can impose different rates. The qualified Quad Cortex path operates at 48 kHz, while the tested JBL Bluetooth path operates at 44.1 kHz.

If a device refuses a configuration or the path behaves differently after a device swap:

1. determine the actual rate required by each device;
2. determine which application/device owns the clock;
3. check whether Matrix Bench is bridging independent output clocks;
4. verify the actual rate used by the measurement application;
5. repeat the reference measurement after the topology is stable.

Do not compare sample counts across different sample rates without converting them to physical time.

10.7 Buffer size did not produce the expected latency

A selected buffer size is not end-to-end latency.

The complete path can include:

- client callback buffering;
- driver safety buffering;
- converter delay;
- DSP;
- asynchronous bridging;
- sample-rate conversion;
- virtual-device buffering;
- wireless buffering.

Verify that the device/client accepted the requested buffer and measure the actual path.

Matrix Bench buffer settings describe application/device buffering, not complete end-to-end latency. Measure the actual path when timing matters.

10.8 Bluetooth feels late

That can be entirely normal.

Bluetooth audio adds buffering outside the application's selected processing buffer. A 16-sample application buffer does not imply a 16-sample Bluetooth path.

If Bluetooth is only being used for monitoring, keep it outside latency-critical conclusions. If Bluetooth itself is the DUT, measure and report the complete wireless path.

10.9 Spectral trace looks unstable

For broadband noise, instantaneous FFT bins naturally fluctuate.

Use averaging and, where appropriate, Noise Smooth to improve visual readability. Peak Hold can help expose excursions.

Do not mistake a smoother display for additional measurement bandwidth or a changed source signal.

For a discrete tone that should be stable, investigate clipping, clocking, DUT modulation, source stability and window/FFT choices instead of merely increasing averaging.

10.10 A spectral peak is not at the expected level

Check the measurement definition before treating the difference as calibration error.

Relevant factors include:

- window coherent gain;
- FFT-bin alignment;
- whether the quantity is amplitude or power;
- analyzer mode;
- Input Gain;
- DUT gain;
- generator Output Level.

Spectral Bench applies its documented window/calibration semantics. Use the defined single-tone measurement mode when the goal is a calibrated tone measurement rather than estimating amplitude visually from an arbitrary spectrum bin.

10.11 THD looks wrong

Confirm that:

- the fundamental is valid and sufficiently above noise;
- the DUT is not clipping;
- the selected mode matches the generated tone;
- the relevant harmonics remain below Nyquist;
- no unrelated tone is being interpreted as DUT distortion.

THD is based on the documented harmonic definition. It is not interchangeable with THD+N.

When comparing devices, keep stimulus frequency, level and measurement conditions identical.

10.12 THD+N looks wrong

THD+N includes noise within the documented measurement bandwidth.

A wider bandwidth can produce a larger THD+N value even when harmonic distortion is unchanged. Preserve the bandwidth with every comparison.

If THD is normal but THD+N is unexpectedly high, investigate analog noise, gain staging, grounding, interface noise and the defined bandwidth before assuming a harmonic-distortion problem.

10.13 IM products appear to be missing

For CCIF/DFD-style and SMPTE-style measurements, verify that Signal Bench and Spectral Bench are using matching defined modes.

Some products may be unavailable when they fall beyond Nyquist.

The suite reports selected products rather than claiming a universal standards-defined aggregate IM percentage. Do not infer a missing aggregate metric from the selective product display.

10.14 Referenced sweep is not flat in direct loopback

A direct dual-channel loopback should be the first diagnostic.

Check:

- reference and DUT channels are not swapped;
- both paths use equivalent cabling/gain;
- no Matrix processing is active on one path;
- interface mixer routing is symmetrical;
- sample rate is stable;
- the DUT really has been removed from the DUT path.

Small residual channel mismatch can be physical. A direct-loopback reference helps show whether the difference belongs to the interface/channel pair rather than to the DUT.

10.15 Raw phase has a large slope

A constant transport delay appears as a phase slope with frequency. This is expected in Raw phase.

Use the phase reference that matches the question:

- Raw to retain complete measured phase;
- Auto to remove a response-dependent estimated constant delay;
- Manual to remove a known independently established delay;
- Baseline for A/B phase comparison.

Do not interpret a Raw delay slope as evidence that the DUT has an unusual frequency-dependent phase response before accounting for transport.

10.16 Auto and Latency Bench disagree

They are not measuring the same quantity in the same way.

Spectral Bench Auto chooses a response-dependent constant-delay compensation for phase display. Latency Bench measures physical relative timing with its deterministic probe/correlation method.

For example, a path can produce a different Spectral Bench Auto compensation value from its independently measured Latency Bench physical latency. Manual phase compensation should use the independently established delay when that is the quantity required.

A numerical difference is therefore not automatically a fault.

10.17 Polarity is reversed

Polarity reversal changes correlation sign, not physical arrival time.

Latency Bench timing selection uses correlation magnitude, so a clean inverted DUT can still produce the same delay.

In Spectral Bench, polarity inversion corresponds to a phase inversion and should be interpreted as part of the transfer response.

Do not "correct" polarity merely to make a latency number appear familiar.

10.18 Latency result is unexpectedly large after filtering

A steep filter can add substantial phase/group-delay timing.

Latency Bench measures complete response arrival through the tested path. For a strongly filtered output, that includes the timing consequence of the filter itself.

The qualified 20-160 Hz cases measured much later than the broadband bypass path, and the change was physically consistent with filter phase/group delay.

If the engineering question is bare transport latency, establish it in a broadband/bypass state where possible. If the question is when the filtered output arrives, the filtered complete-response result is the relevant one.

10.19 Latency result is rejected as ambiguous

Do not force a number.

Narrow or strongly periodic responses can produce multiple plausible correlation peaks. Latency Bench rejects a result when a sufficiently strong separate candidate competes with the primary candidate.

Try:

1. verify level and clipping;
2. verify routing and baseline;
3. repeat the measurement;
4. use the legitimate DUT bandwidth, not an artificially widened path;
5. allow automatic extended analysis to run;
6. if it still rejects, report that the available response does not support a unique timing result under those conditions.

Rejection is preferable to a precise-looking arbitrary peak.

10.20 Extended latency analysis was used

Extended analysis is automatic. Its use does not by itself mean the result is poor.

It is intended for paths where the normal broadband probe does not provide sufficiently robust timing evidence, including strongly bandwidth-limited outputs.

Extended analysis uses a longer probe/correlation window and magnitude-matches the captured reference spectrum to the DUT spectrum while preserving DUT phase.

Record that extended analysis was used when documenting the result.

10.21 Baseline is stale

A Latency Bench baseline belongs to the setup that created it.

Re-establish it after changes such as:

- interface;
- sample rate;
- channels;
- physical cabling;
- reference/DUT loop assignment;
- other changes that alter the fixed path mismatch.

If a difficult DUT result looks suspicious, return to direct/bypass and verify the reference before continuing.

10.22 Latency varies between runs

Inspect the reported spread.

Small sub-sample variation can be normal. Large variation can indicate:

- unstable routing;
- changing DUT state;
- asynchronous behavior;
- weak correlation evidence;
- clipping/noise;
- wireless buffering;
- clock-domain effects.

Do not hide variation by reporting only one selected run. Use the median and accompanying statistics.

10.23 Matrix device is selected but there is no audio

Device selection and routing are separate.

Check:

- input device/channel;
- required crosspoint;
- crosspoint gain;
- Main/Aux destination device/channel;
- destination gain;
- mute;
- input filters/polarity;
- compressor/bypass if relevant.

A selected Main or Aux output receives nothing until a crosspoint routes an input to it.

10.24 Matrix route changed after a device swap

Channel identity matters.

If a selected channel exists on both devices, Matrix Bench preserves the valid channel identity rather than intentionally collapsing every device to a generic first stereo pair.

Verify the actual available channel indexes on both devices and inspect the effective routing after the swap.

Do not rebuild the matrix until you know whether the observed behavior is a valid preserved mapping or an actual routing fault.

10.25 Matrix device disappeared

For a temporary disconnect, preserve the intended logical configuration.

MatrixBenchEngine is designed to retain device identity and recover the route when the device returns.

If recovery fails:

1. verify the device is visible to macOS;
2. verify its sample rate/availability;
3. verify the engine is running;

4. reopen Matrix Bench and inspect the retained device/channel state;
5. only then rebuild routing if necessary.

10.26 Matrix audio stops when the window closes

Established Matrix Bench routing should continue when the application window is closed.

If audio stops, reopen Matrix Bench and verify that the intended devices and routes are still present. If the background routing component is unavailable, restart Matrix Bench or the Mac before rebuilding the routing configuration.

Treat this as an application/runtime problem rather than as expected behavior from closing the window.

10.27 Matrix Bench opens but routing state is unavailable

If Matrix Bench opens without its expected routing state, first verify that the installed application is functioning normally and that the relevant audio devices are available.

Close and reopen Matrix Bench. If the problem remains, restart the Mac before recreating routes or snapshots; this avoids replacing a valid saved configuration because of a temporary runtime failure.

Do not repeatedly launch additional copies of the application as a substitute for restoring the normal routing state.

10.28 Virtual Matrix route has unexpected channels

The Matrix virtual device is 8-in/8-out and can be used as multichannel audio or stereo pairs.

Verify channel selection in both Matrix Bench and the receiving/sending macOS application. For OS-side mappings, also inspect Audio MIDI Setup or the relevant macOS channel configuration.

Non-adjacent channel mappings can be valid. Do not assume every application uses only channels 1-2.

10.29 Virtual routing creates feedback

Treat the virtual path exactly like physical patch cables.

A loop can occur when an application's output enters Matrix Bench and Matrix Bench routes it back to an input that the same application is monitoring or retransmitting.

Mute or remove one edge of the loop before increasing gain.

Matrix Bench also rejects specific unsafe simultaneous virtual Main/Aux configurations as a feedback-prevention measure. A rejection message can therefore indicate correct protection rather than device failure.

10.30 Independent Matrix outputs drift or behave differently

Main and Aux can target independent physical devices with independent clocks.

When clocks differ, Matrix Bench uses the architecture required to bridge those output domains. The path is no longer equivalent to one interface running all channels from one clock.

For precision measurements, simplify to one clock domain when possible. If independent devices are part of the system under test, document them as such. Matrix Bench's independent-output path uses asynchronous sample-rate conversion so Main and Aux can remain active on devices whose clocks are not locked to each other.

10.31 Matrix snapshot recall sounds wrong

First verify the snapshot contents rather than repeatedly recalling it.

Snapshots can restore coordinated routing, mutes, polarity, filters and gains. A saved state can therefore be consistently wrong if it was captured incorrectly.

Check the named snapshot against the intended:

- devices/channels
- crosspoints
- gains
- mutes
- INV
- HPF/LPF
- destination state

Atomic recall minimizes audible intermediate states; it cannot make an incorrectly stored state correct.

10.32 Matrix compressor changes a measurement

Main and Aux master compressors are real processing.

If the purpose is to measure the external DUT without Matrix dynamics processing, bypass the compressor on the measurement destination.

Also inspect makeup gain and gain-reduction meters. A compressor can change level, spectrum and apparent DUT behavior even when routing itself is correct.

10.33 MIDI Bench shows no incoming messages

Check:

1. physical MIDI connection;
2. MIDI IN device;
3. channel;
4. whether the controller is actually transmitting;
5. Raw bytes.

Raw-byte display helps distinguish "no traffic" from "traffic exists but is not the message type expected."

If available, compare with the deterministic test sender to separate MIDI Bench behavior from the physical controller.

10.34 MIDI target does not respond

Verify MIDI OUT, channel and message type.

For CC, verify controller number and value. For PC, verify the actual Program Change value and the receiving device's numbering convention.

If a physical controller operates the target, capture that known-working message in MIDI Bench and reproduce it manually. This isolates the target from the controller.

10.35 Program number appears off by one

MIDI program numbering and front-panel preset numbering are not universally displayed with the same convention.

Verify the actual MIDI Program Change value before adding or subtracting one.

Document the MIDI value and the DUT's displayed preset separately when ambiguity matters.

10.36 Command-file Run is disabled

The selected MIDI Bench command file is invalid.

Correct the syntax and allow the application to revalidate it. Run should become available again when the file is valid.

Do not work around validation by trying to execute only the lines that "look right." Deterministic bench control depends on the whole sequence being understood.

10.37 MIDI loop changes the DUT too quickly

Add explicit WAIT delays.

A MIDI message can be transmitted quickly while the hardware needs substantially longer to switch presets, relays, converters or DSP state.

For measurement work, settling time belongs to the test definition.

10.38 MIDI state and audio state do not line up in time

MIDI Bench timestamps the message activity; it does not prove when the DUT's audio path finished changing.

Do not assume the MIDI timestamp is the audio transition timestamp.

If the timing relationship matters, measure it separately or use a conservative known settling interval before capturing the audio result.

10.39 Signal Bench output is unexpectedly quiet

Check Output Level and any active broadband filtering before increasing downstream gain.

Also inspect:

- selected signal mode;
- Dual Sine individual tone levels;
- mute;
- slicer state;
- Pick Attack behavior where applicable;
- interface output level.

Noise RMS is intentionally below peak/full scale. Do not expect Pink or White noise to show the same RMS as a full-scale sine.

10.40 Pink-noise mean is not exactly zero

A finite Pink-noise record can have a non-zero mean because very slow components may not average exactly to zero within one observation window.

Judge Pink noise primarily by its intended spectral and RMS behavior rather than requiring one finite capture to have exactly zero mean.

10.41 A result changed after reopening an application

Check persisted state before assuming nondeterminism.

The Bench applications intentionally persist relevant configuration. A reopened tool can therefore return to a previous device, channel, gain or mode rather than a generic default.

For a formal measurement, record the state explicitly and verify it at the start of the session.

10.42 A result is too perfect

Treat implausibly perfect results with the same suspicion as obviously bad ones.

Examples include:

- exactly zero phase where a physical path should contain delay;
- zero run-to-run variation after a topology change;
- an impossibly flat DUT response;
- distortion below the known measurement floor;
- latency equal to a selected buffer size by coincidence.

Check whether the measurement is accidentally observing the reference path, a bypass route, a virtual shortcut or the wrong channel.

A believable number is not evidence that the intended DUT was measured.

10.43 Two tools give different numbers

First ask whether they are defining the same quantity.

Examples:

- Spectral Auto phase delay versus Latency Bench physical latency;
- filtered complete-response timing versus broadband transport timing;
- spectrum-bin level versus calibrated single-tone measurement;
- THD versus THD+N;
- selected buffer size versus physical round-trip latency.

Resolve the definitions before treating the difference as an application fault.

10.44 After a surprising result

Use this recovery sequence:

1. Save the surprising result and setup notes.
2. Do not immediately "fix" the DUT or measurement settings.
3. Return to the simplest known reference.
4. Verify devices, channels, sample rate and levels.
5. Repeat the reference.
6. Reproduce the DUT state with one controlled change.

7. Repeat the measurement.
8. Compare the complete evidence, not only the headline number.

Preserving the first surprising result is useful. It may contain the evidence needed to distinguish a real DUT behavior from a setup error.

10.45 When to stop troubleshooting

Stop trying to force the expected answer when:

- the instrument correctly rejects the measurement;
- the physical path cannot support the required bandwidth or level;
- the DUT behavior is genuinely state-dependent;
- an independent clock or wireless path is intrinsically variable;
- the question being asked is not the quantity the selected Bench measures.

At that point, redefine the experiment or report the limitation. Measurement engineering includes knowing when the available evidence does not support a stronger conclusion.

11. Technical reference

This appendix collects compact technical reference material for users. Application-specific operating details remain in Chapters 3 through 7, cross-Bench procedures in Chapter 8, and troubleshooting in Chapter 10.

11.1 Units and conventions

Samples and time

At sample rate F_s , one sample represents:

$$\text{sample time} = 1 / F_s \text{ seconds}$$

and a delay of N samples represents:

$$\text{delay_ms} = 1000 * N / F_s$$

Examples:

Sample rate	One sample
44.1 kHz	0.022676 ms
48 kHz	0.020833 ms
96 kHz	0.010417 ms
192 kHz	0.005208 ms

Fractional-sample timing estimates are valid when the estimator supports sub-sample interpolation. They do not imply that the audio hardware processes fractional physical samples.

Frequency

Frequency is expressed in hertz (Hz) or kilohertz (kHz). Unless a specific logarithmic relationship is stated, frequency values are ordinary physical frequency rather than FFT-bin numbers.

Level

Digital audio level is normally expressed in dBFS. 0 dBFS is the digital full-scale limit for the represented quantity.

The handbook does not use dBFS as a universal substitute for analog voltage, SPL or power. Conversion to an analog quantity requires the relevant calibration.

Polarity and phase

Polarity is sign inversion.

Phase is frequency-dependent angular relationship.

A polarity inversion corresponds to 180 degrees for a sinusoidal component, but polarity and arbitrary phase response are not interchangeable concepts.

11.2 Core terminology

Actual

Spectral Bench sweep result representing the captured DUT response rather than the reference-normalized transfer response.

Baseline

A stored reference measurement or state used to remove fixed path mismatch or compare against a later state. The exact semantics depend on the Bench application.

Buffer size

Number of audio frames requested or used for a processing callback/buffer. It is not synonymous with complete path latency.

CC

MIDI Control Change.

Complete-response timing

Arrival timing of the measured DUT response, including timing effects caused by its transfer function, such as filter phase/group delay.

Correlation

Similarity measure used by Latency Bench to estimate relative timing between reference and DUT captures.

dBFS

Decibels relative to digital full scale.

DUT

Device under test.

Extended analysis

Latency Bench fallback analysis for difficult responses. It uses a longer probe/correlation window and magnitude-matches the reference spectrum to the captured DUT spectrum while retaining DUT phase.

FFT

Fast Fourier Transform.

Group delay

Frequency-dependent timing associated with phase response.

HAL

macOS Core Audio Hardware Abstraction Layer.

Main / Aux

First-class Matrix Bench output destinations with independent routing, gain, mute and master compression.

PC

MIDI Program Change.

Reference path

Known or simultaneously measured path used to establish the comparison state.

Selection evidence

Latency Bench timing-candidate score based on correlation magnitude and actual overlap. It is distinct from the raw reported correlation.

THD

Total harmonic distortion under the documented Spectral Bench harmonic definition.

THD+N

Total harmonic distortion plus noise under the documented Spectral Bench definition and measurement bandwidth.

Transfer

Spectral Bench referenced sweep result representing DUT path relative to the simultaneously captured reference path.

11.3 Current macOS handbook versions

This handbook edition documents the current qualified macOS releases of the five Bench applications.

Application	macOS version
Matrix Bench	2.1.1
MIDI Bench	2.0.0
Signal Bench	2.0.0
Spectral Bench	2.1.0

Latency Bench	1.1.1
---------------	-------

Version numbers can differ between Bench applications; record the version used for any measurement that may need to be reproduced later.

This edition is macOS-first. Other platforms are outside the scope of the operating procedures unless explicitly noted.

11.4 macOS suite location

The intended application-suite location is:

/Applications/60°N Signal Works Audio Bench Suite

Installation packaging can change independently of the measurement method. Follow the package instructions supplied with the version you are using.

Where a package version is stated, it refers to the qualified artifact documented by the corresponding application chapter.

11.5 Signal Bench quick reference

Signal Bench 2.0.0 provides:

- Pink noise
- White noise
- Dual Sine
- broadband HPF / LPF
- defined measurement presets
- guitar interval/tuning functions
- harmonic enrichment
- Slicer
- Pick Attack
- Output Level
- Mute
- final output metering

For measurement work, record the signal mode, frequencies, individual tone levels where applicable, overall Output Level, filter settings and any modulation/enrichment state.

Signal Bench does not insert a protective output limiter. Headroom remains the user's responsibility.

11.6 Spectral Bench quick reference

Spectral Bench 2.1.0 includes:

- FFT sizes: 4096 / 8192 / 16384 / 32768 / 65536
- Windows: Hann / Blackman-Harris 4-term / Flat Top
- Averaging: Off / Fast / Medium / Slow
- Peak Hold
- Raw / Noise Smooth spectrum presentation
- Input Gain
- frequency and dB viewport controls
- single-tone measurements

H2-H10
THD
THD+N
CCIF / DFD-style selective products
SMPTE-style selective products
referenced sweeps
Actual / Transfer
Raw / Auto / Manual / Baseline phase
PNG / CSV / TXT saved results

Input Gain modifies analyzer input samples. Viewport controls modify presentation only.

For saved results, preserve the analyzer/sweep mode and phase-compensation metadata rather than inferring them later from the plotted image.

11.7 Latency Bench quick reference

Latency Bench 1.1.1 uses a two-path physical measurement:

Reference OUT A -> direct/reference path -> IN A
DUT OUT B -> DUT path -> IN B

The baseline measures the fixed relative mismatch of the direct dual-loop setup. DUT measurement then reports DUT-path timing relative to that baseline.

Normal-analysis constants:

probe length:	2048 samples
capture length:	32768 samples
prime:	2048 samples
tail:	12288 samples
maximum signed lag:	16384 samples
correlation window:	4096 samples
default probe level:	-30 dBFS

Extended analysis uses:

probe length:	8192 samples
correlation window:	12288 samples

The selection score is conceptually:

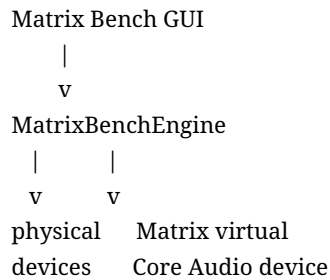
$$E = |r| \sqrt{(N_o / N_w)}$$

A valid primary candidate requires adequate evidence. A sufficiently strong separate candidate at or above the documented 90% competing-evidence ratio causes rejection.

Timing selection is based on correlation magnitude, so polarity inversion does not by itself change the measured arrival time.

11.8 Matrix Bench quick reference

Matrix Bench 2.1.1 architecture:



Core signal flow:

input
 -> raw input meter
 -> INV
 -> HPF / LPF
 -> matrix crosspoint gain
 -> Main or Aux destination gain/mute
 -> destination master compressor
 -> output

Qualified selectable buffer sizes are:

16 / 32 / 64 / 128 / 256 / 512

The Matrix virtual device provides 8 inputs and 8 outputs to macOS audio clients. Multichannel-aware software can address all eight channels; ordinary stereo use can use pairs 1-2, 3-4, 5-6 or 7-8.

The persistent engine service label is:

works.60n.matrixbench.engine

The GUI installs/starts that service as needed; the audio engine is not owned by the GUI process.

The GUI is a control surface for the persistent engine; closing it should not stop qualified headless routing.

matrixbenchctl exists as an engineering/service command-line interface, not as a prerequisite for normal operation.

11.9 MIDI Bench quick reference

MIDI Bench 2.0.0 provides:

MIDI IN selection
 MIDI OUT selection
 channel selection
 Incoming monitor
 Outgoing monitor
 timestamps
 Raw bytes
 Pause
 Auto-scroll
 Clear
 manual CC
 manual PC
 Run from file
 Browse / Edit / Run / Stop / Syntax

WAIT delay commands
persistent device/channel state

Examples of the documented command-file forms:

CH 1 CC 12 VAL 67
CH 3 PC 4

The project README remains authoritative for the exact accepted delay-command grammar and parser rules.

11.10 Cross-Bench quantity ownership

Use the tool whose measurement definition matches the quantity:

Quantity	Primary Bench
stimulus definition	Signal Bench
level/spectrum/harmonics	Spectral Bench
THD / THD+N / selective IM products	Spectral Bench
referenced magnitude response	Spectral Bench
referenced phase response	Spectral Bench
physical-path arrival timing	Latency Bench
complete-response timing of a filtered path	Latency Bench
audio routing / processing state	Matrix Bench
MIDI traffic / deterministic MIDI command state	MIDI Bench

This table prevents similar-looking values from being treated as interchangeable measurements.

11.11 Latency interpretation reference

Buffer latency

A buffer contributes time proportional to its frame count and sample rate, but the complete path can contain several buffers and other fixed/variable delays.

Transport latency

Transport latency describes the path's underlying propagation/processing delay without intentionally attributing filter group-delay behavior to transport.

Complete-response timing

Latency Bench measures the arrival supported by the actual response presented to it. For a strongly filtered path, the result can therefore include substantial filter phase/group-delay timing.

Acoustic propagation

At ordinary room conditions, sound in air takes roughly 2.9 ms per metre. Acoustic distance can therefore dominate electronic low-latency differences.

When microphones and loudspeakers are part of a latency experiment, include their physical spacing in the interpretation.

11.12 Phase-reference summary

Spectral phase mode	Meaning
Raw	measured phase including transport delay
Auto	phase after response-dependent estimated constant-delay removal
Manual	phase after user-specified known delay removal
Baseline	phase relative to a stored comparison state

Use Manual when an independent Latency Bench result is intentionally being used as the constant-delay reference.

Do not assume Auto is an independent physical-latency measurement.

11.13 Distortion-reference summary

THD is based on the documented harmonic set relative to the fundamental.

THD+N includes non-fundamental content/noise within its defined bandwidth.

CCIF / DFD-style and **SMPTE-style** modes report selected intermodulation products defined by Spectral Bench. They are not silently converted into a standards-defined aggregate IM percentage.

Every comparison should preserve frequency, level, sample rate and measurement bandwidth where applicable.

11.14 Matrix routing record

For a reproducible Matrix configuration, record:

- Matrix Bench version
- input device / channels
- Main device / channels
- Aux device / channels
- sample rate
- buffer size
- crosspoint gains
- input INV / HPF / LPF
- Main/Aux gain and mute
- compressor state/settings
- snapshot name

MIDI assignments where relevant
virtual-channel mapping
physical wiring

For latency-sensitive routing, add the measured path latency rather than deriving it from the selected buffer.

11.15 Measurement record template

A compact generic record can use:

Title:
Date:
Purpose / question:

Applications and versions:
Audio interface/device:
Sample rate:
Buffer size:
Physical wiring:
Virtual routing:

Stimulus:
Stimulus level:
DUT:
DUT state:
MIDI state/command:

Reference/baseline:
Measurement mode:
Bandwidth:
Phase compensation:

Primary result:
Run statistics/repeatability:
Acceptance/rejection status:

Saved files:
Notes:

Not every field applies to every test. Omit irrelevant fields rather than filling them with guessed values.

11.16 MIDI test record

For a reproducible MIDI-side test:

MIDI Bench version
MIDI IN device
MIDI OUT device
channel
message type
CC / PC number
value

command-file contents/revision
delay settings
Loop state
target device
target function
observed timestamps

When MIDI controls an audio measurement, keep this record with the corresponding audio result.

11.17 Saved-result file discipline

Prefer descriptive filenames that identify the DUT and state without requiring the containing directory to explain them.

For example:

qc-loop-off-48k.txt
qc-loop-on-48k.txt
sub-lpf-20-160-24db.csv
sub-lpf-20-160-24db.png
matrix-main-aux-routing-notes.txt

Keep companion files from the same measurement together. Do not rename only one member of a Spectral Bench PNG/CSV/TXT result set in a way that obscures their relationship.

For formal comparison, store a short setup record beside the machine-generated files.

11.18 Measurement-session checklist

- application and version;
- audio/MIDI devices and channels;
- sample rate and buffer size where relevant;
- physical or virtual routing;
- stimulus and level;
- DUT state;
- reference or baseline state;
- measurement mode and relevant bandwidth;
- result and repeatability/statistics;
- saved result files and any notes needed to reproduce the setup.

12. Bugs, errata and current packages

This handbook and the Bench applications are engineering tools and may contain bugs, documentation errors or omissions. If you find behavior that appears incorrect, a reproducible application problem, or an error or ambiguity in this handbook, please inform the creator so it can be checked and corrected in a future release.

When reporting a possible bug or erratum, include enough context to reproduce or locate it: the Bench and version, macOS version where relevant, audio/MIDI device and routing, sample rate and buffer size where relevant, DUT state, the steps that produced the issue, and the handbook section/page if the issue is documentation-related.

Before treating an unexpected measurement as an application bug, preserve the result and setup information and check the relevant troubleshooting guidance in Chapter 10. Unexpected results are useful evidence when the original conditions can be reproduced.

Current public installation packages: [Google Drive distribution folder](#).

Because packages and documentation can be updated independently, record the application version used for measurements that need to remain reproducible.